

CS699

Introduction
to
Time Series Analysis

Introduction

- Time series: *An ordered sequence of values of a variable at equally spaced time intervals.*
- Usage: The usage of time series models is twofold
 - Descriptive: Obtain an understanding of the underlying forces and structure that produced the observed data.
 - Predictive: Fit a model and use it to forecast future values.

Introduction

- Applications:
 - Economic Forecasting
 - Sales Forecasting
 - Budgetary Analysis
 - Stock Market Analysis
 - Inventory Studies
 - Workload Projections
 - Utility Studies
 - Census Analysis
 - and many more ...

Introduction

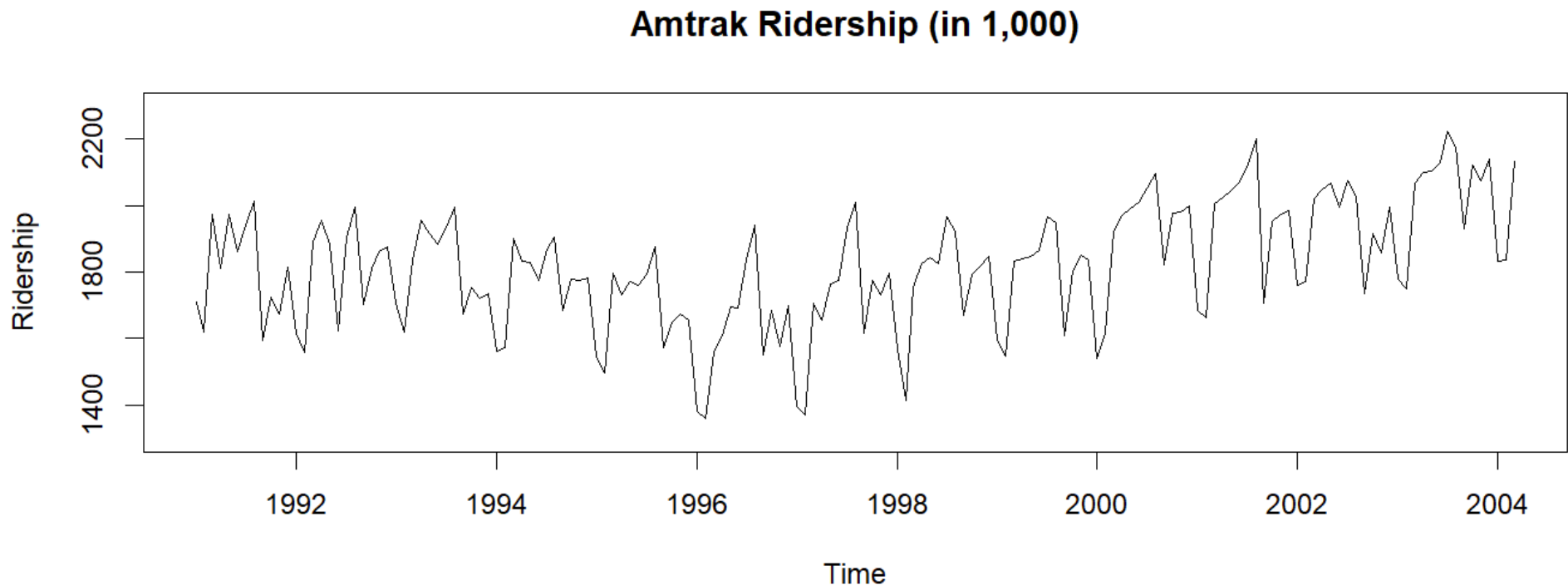
- Forecasting Methods:
 - Regression-based methods
 - Smoothing methods
 - Hybrid
- Will briefly discuss regression-based methods and smoothing methods.

Introduction

- Time Series Components
 - Level: average value of the time series
 - Trend: general direction of the change of values over time
 - Seasonality: identical patterns are observed during corresponding seasons of successive years
 - Noise: random variations or sporadic changes due to measurement errors or chance events
 - Level, trend and seasonality represent underlying characteristics of a time series.

Introduction - Examples

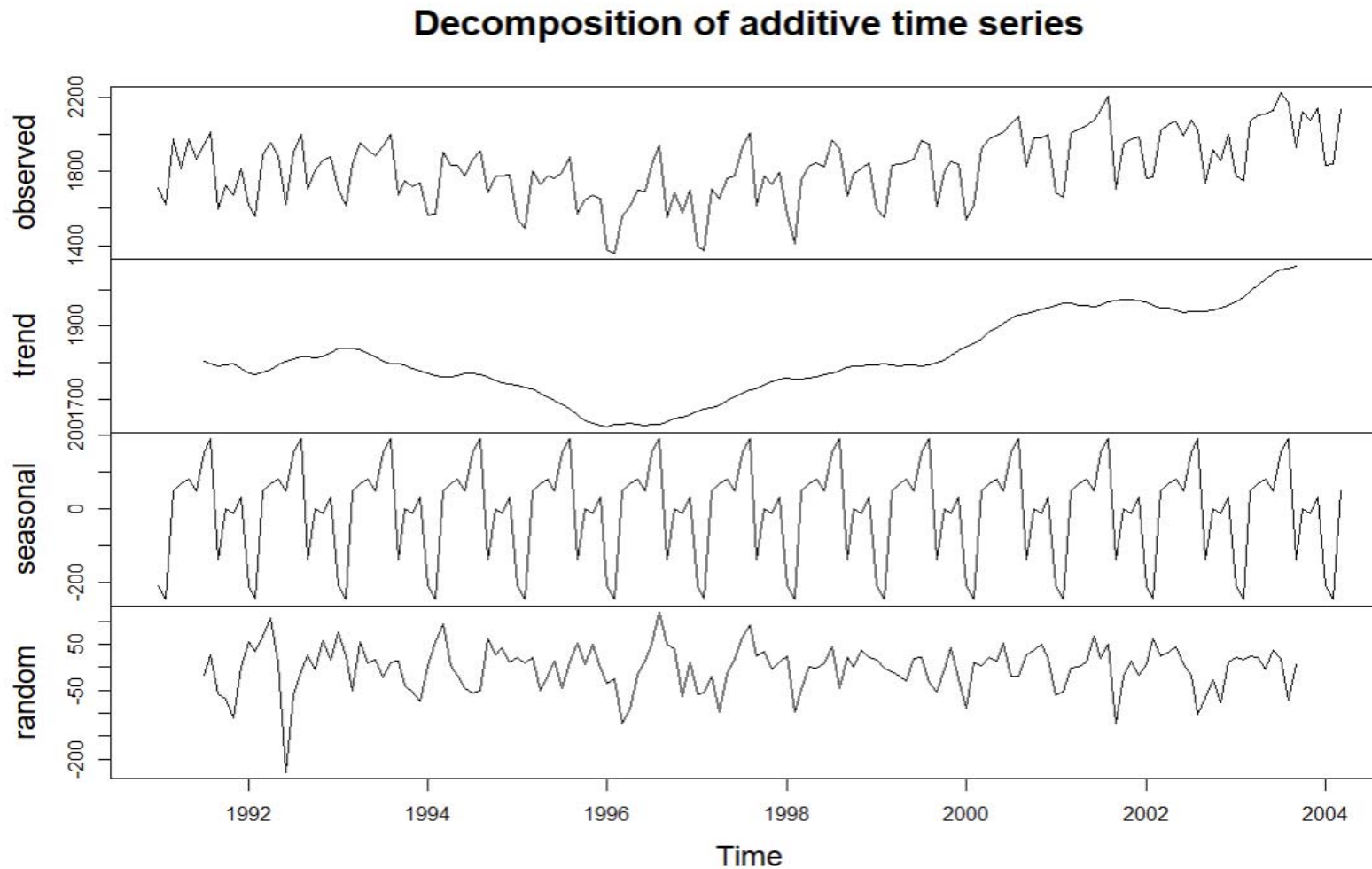
- Amtrak monthly ridership time plot (Jan 1990 – Mar 2004)



- Overall level (average) around 1,800,000
- Slight U shaped trend
- Annual seasonality

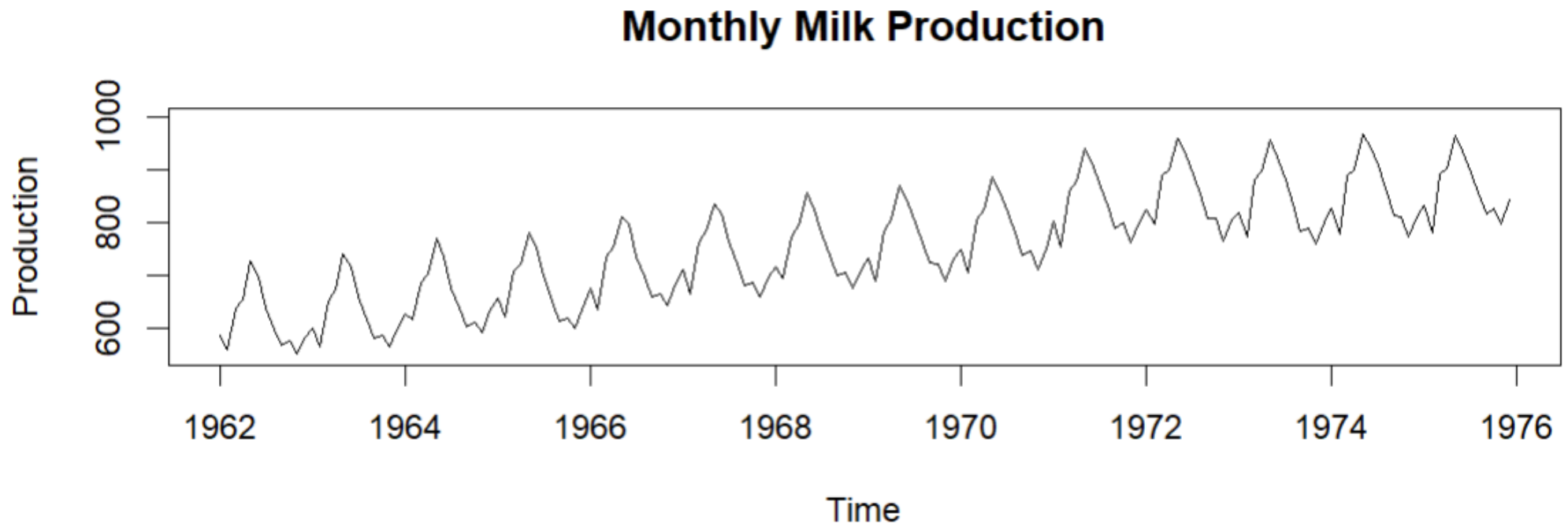
Introduction - Examples

- Amtrak monthly ridership components
 - R's *decompose* function separates four components:



Introduction - Examples

- Monthly milk production time plot, pounds per cow (Jan 1962 - Dec 1975)

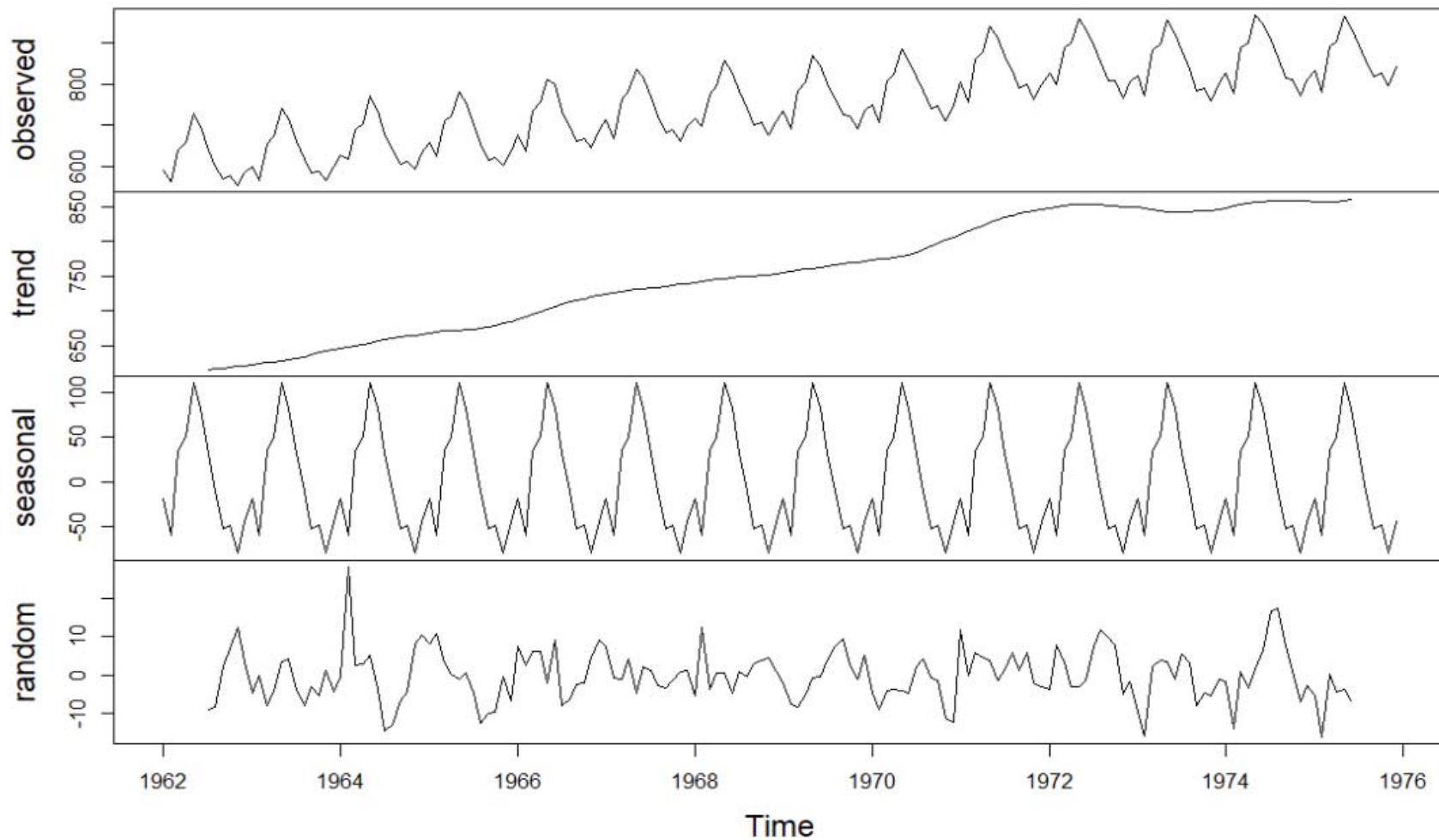


- Overall level: around 750
- Overall trend: increasing
- Annual seasonality

Introduction - Examples

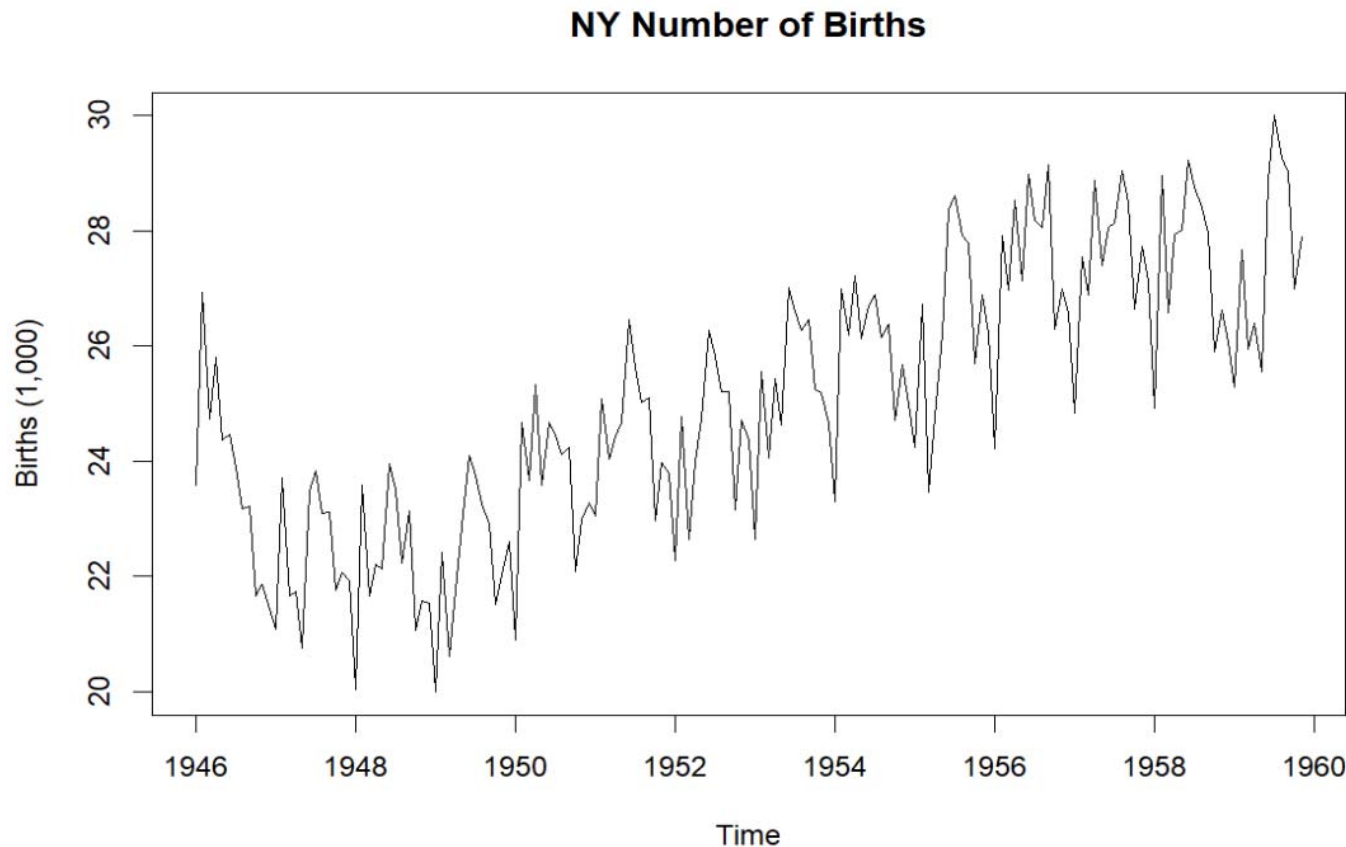
- Monthly milk production components

Decomposition of additive time series



Introduction - Examples

- Number of births per month in NY time plot (1946 – 1959)

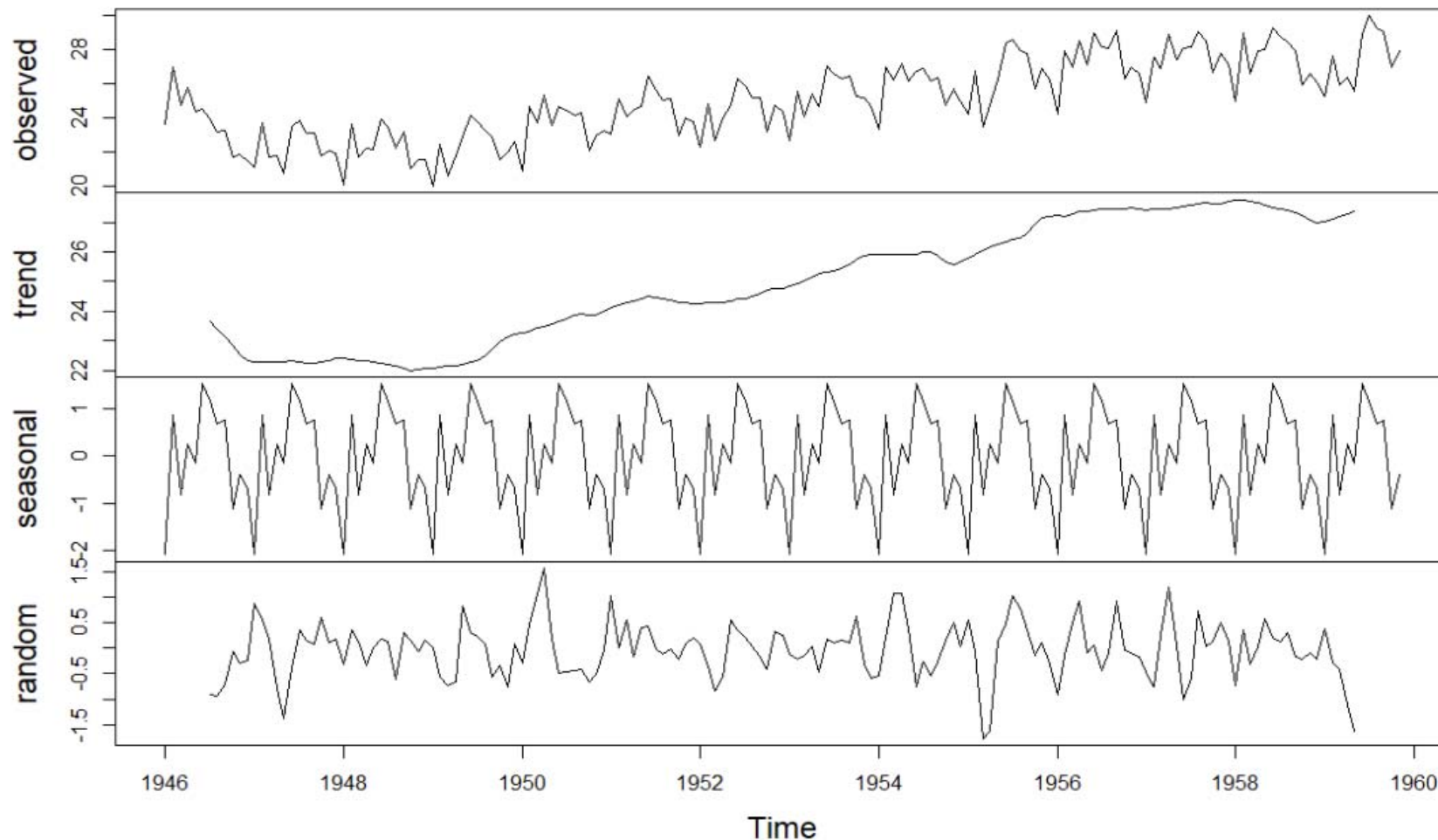


- Overall level: around 25,000
- Overall trend: Increasing
- No obvious seasonality

Introduction - Examples

- Number of births per month in NY components

Decomposition of additive time series



Seasonality

- Detecting seasonality:
 - Time plot
 - Seasonal subseries plot
 - Multiple boxplots
 - Autocorrelation plot

Seasonality

- Seasonal subseries plots are formed by:
 - Vertical axis: output variable
 - Horizontal axis: Time ordered by season. For example, with monthly data, all the January values are plotted (in chronological order), then all the February values, and so on.
- This plot is only useful if the period of the seasonality is already known. In many cases, this will in fact be known. For example, monthly data typically has a period of 12.

Example: CO₂ Concentration

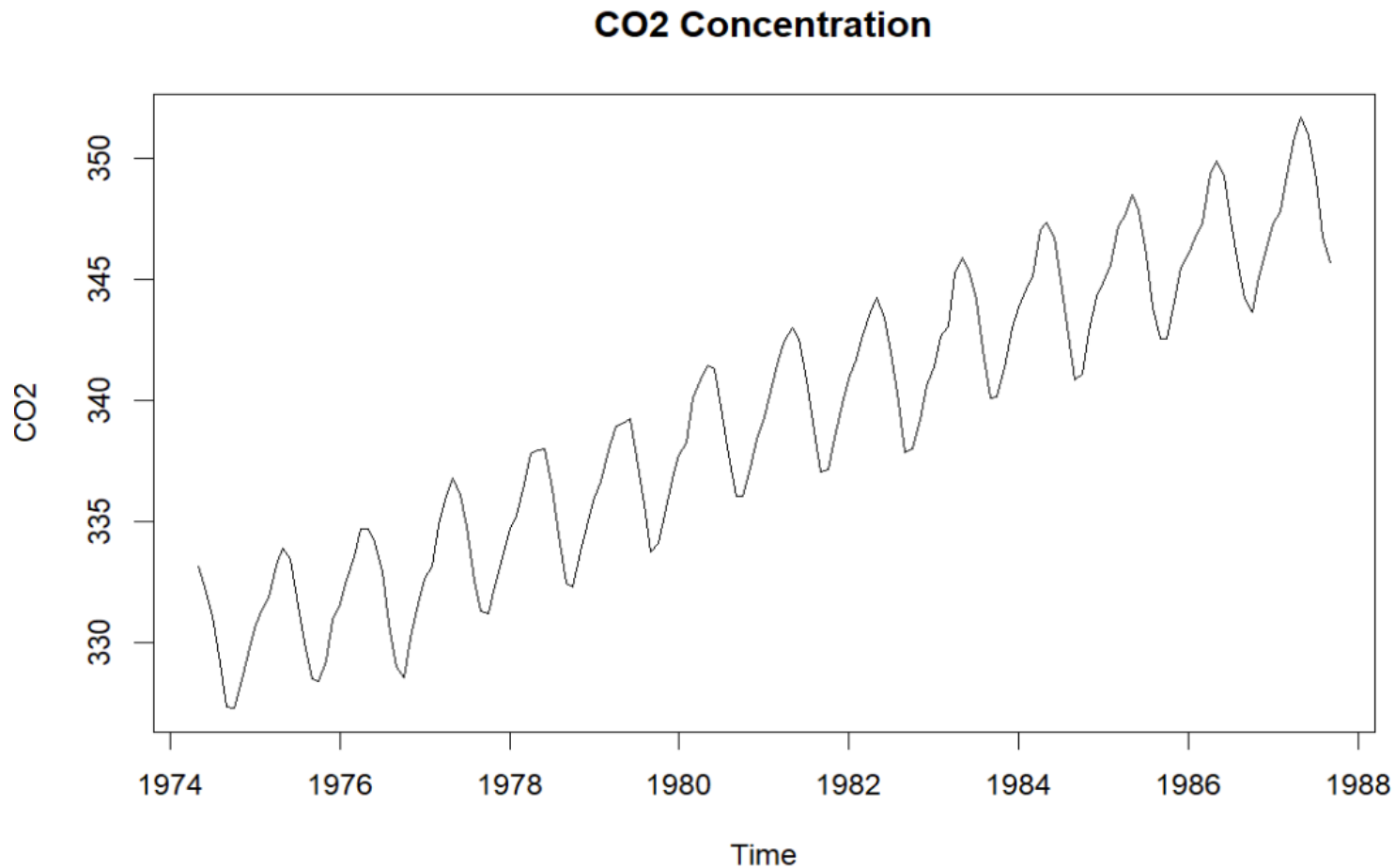
- This data set contains selected monthly mean CO₂ concentrations at the Mauna Loa Observatory from 1974 to 1987.
- Dataset (part)

CO2	Year&Month	Year	Month
333.13	1974.38	1974	5
332.09	1974.46	1974	6
331.1	1974.54	1974	7
329.14	1974.63	1974	8
327.36	1974.71	1974	9
327.29	1974.79	1974	10
328.23	1974.88	1974	11
329.55	1974.96	1974	12

- co2-concentration.csv

Example: CO2 Concentration

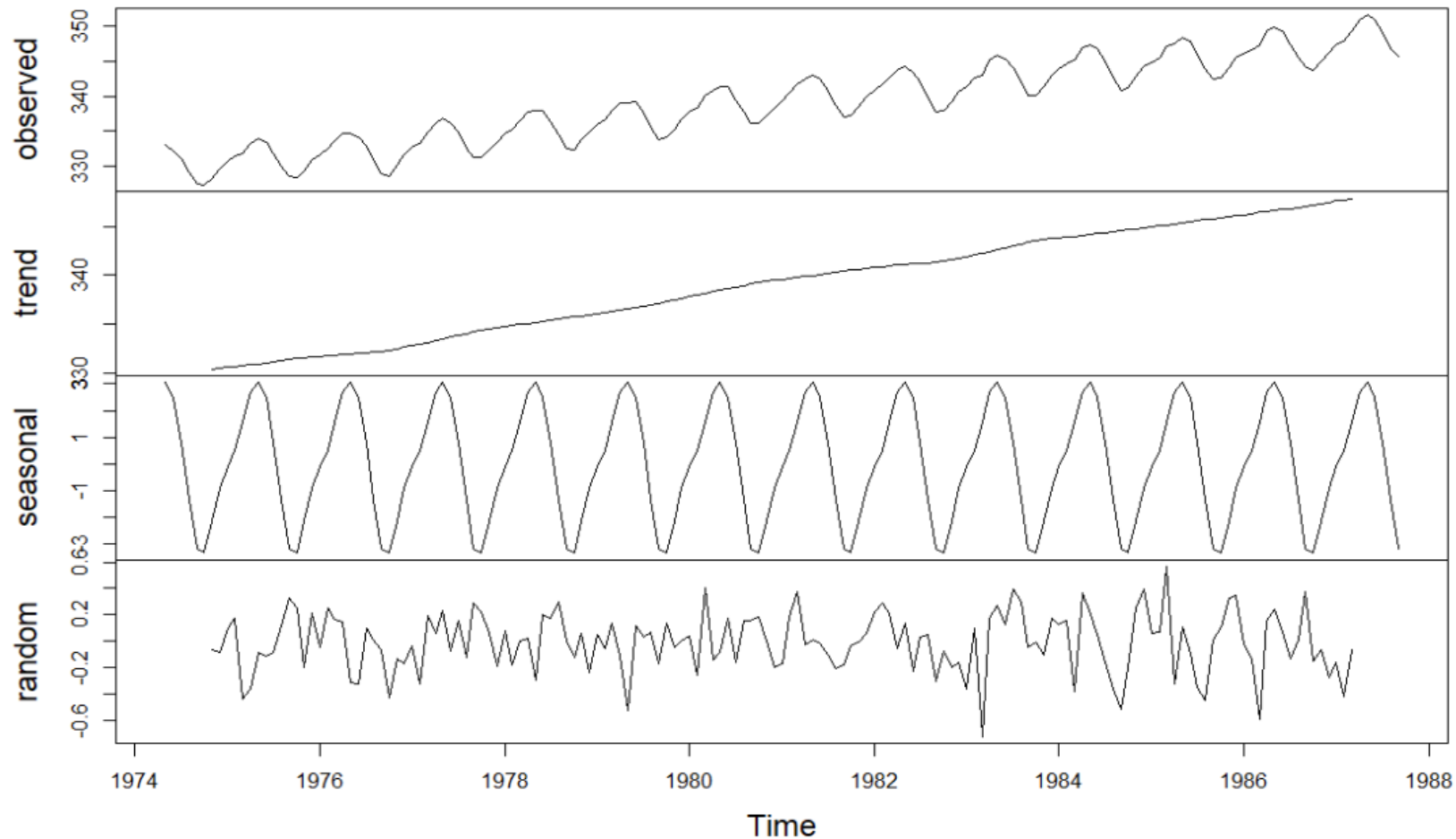
- Time plot



- Can observe a liner increasing trend.

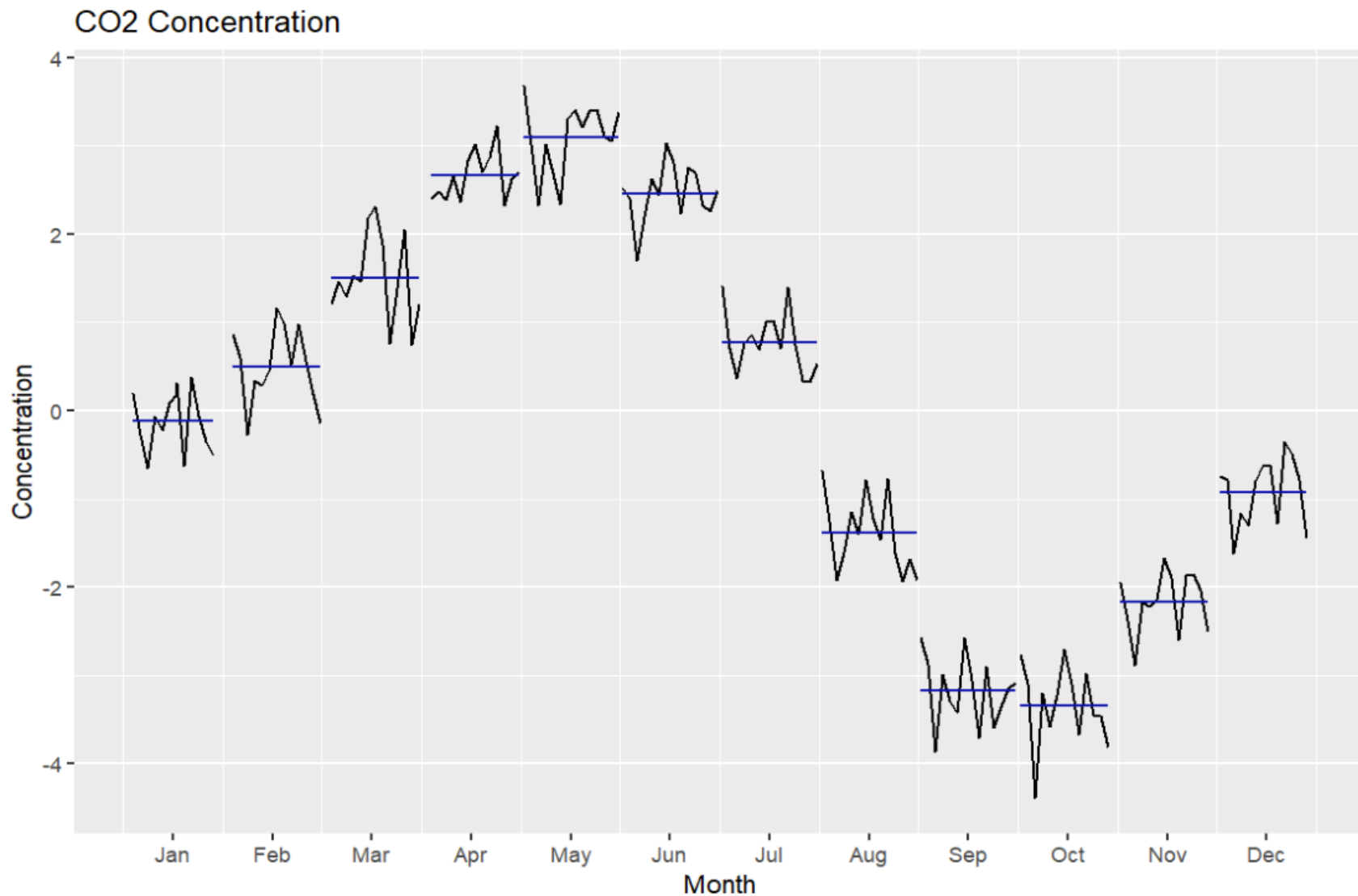
Example: CO2 Concentration

- CO2 concentration components



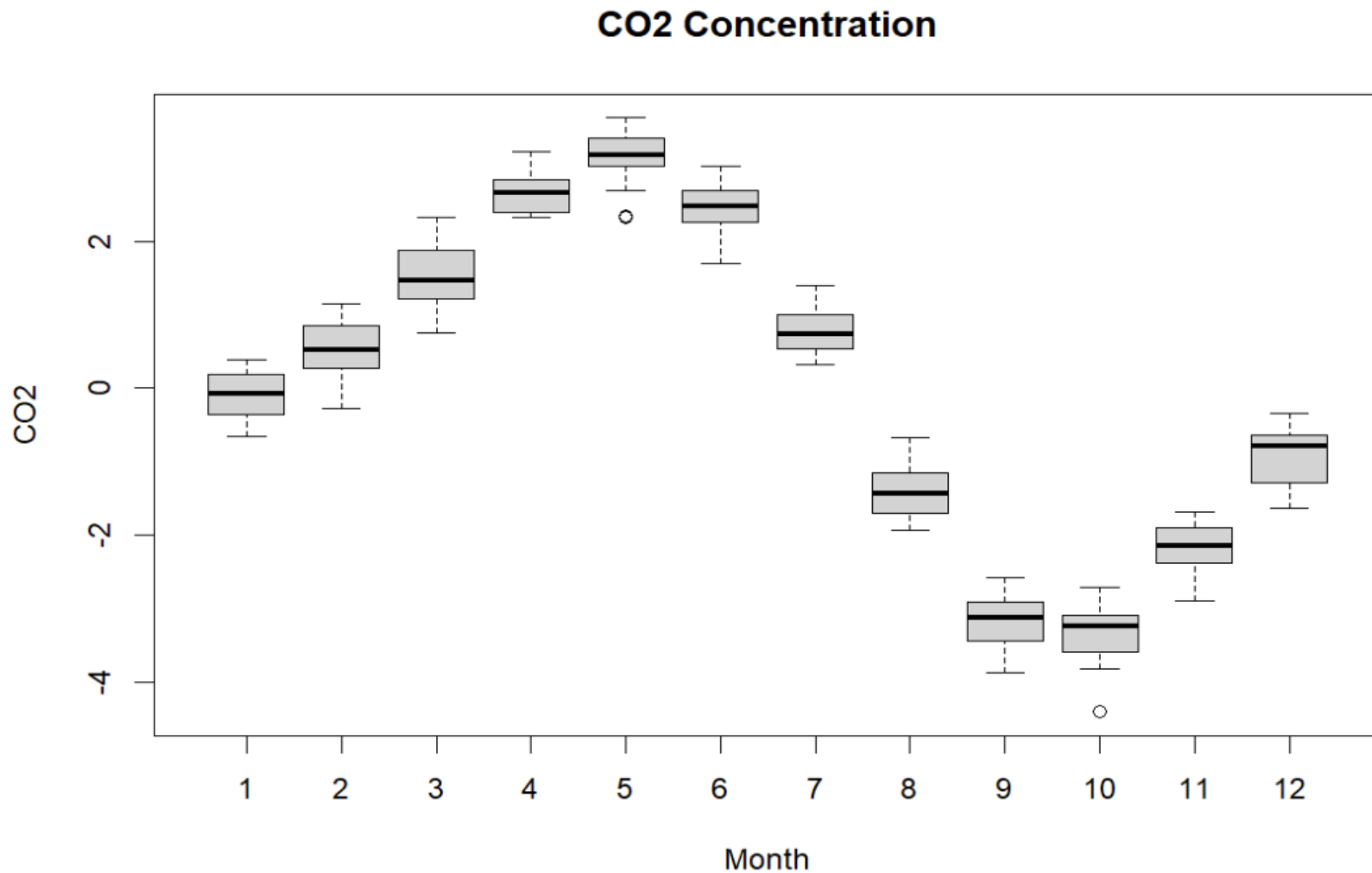
Example: CO2 Concentration

- Seasonal subseries plot



Example: CO2 Concentration

- Seasonal boxplot



Example: Southern Oscillation

- The southern oscillation is defined as the barometric pressure difference between Tahiti and the Darwin Islands at sea level.
- The southern oscillation is a predictor of El Nino which in turn is thought to be a driver of world-wide weather.
- Specifically, repeated southern oscillation values less than -1 typically defines an El Nino.

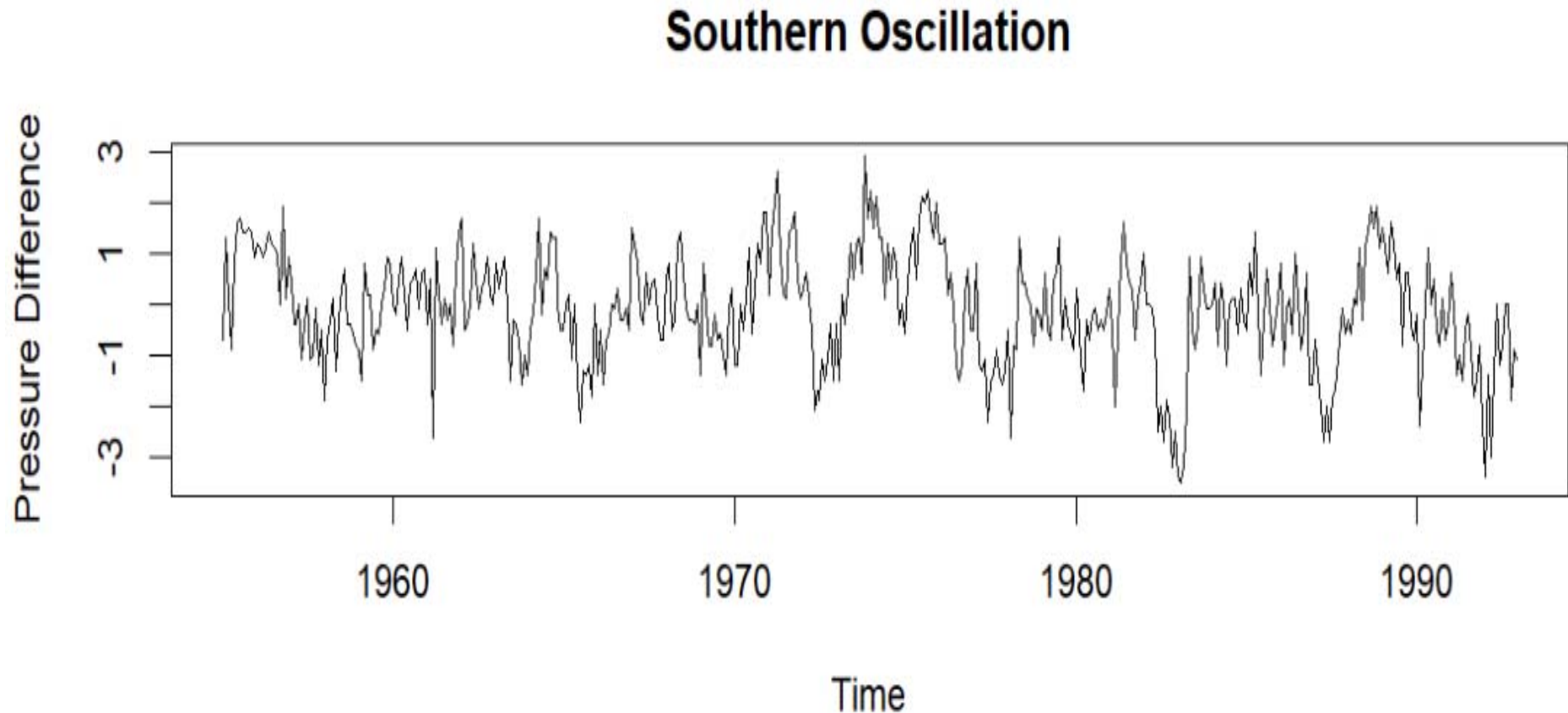
Example: Southern Oscillation

- Data (part)

Oscillation	Year_fraction	Year	Month
-0.7	1955.04	1955	1
1.3	1955.13	1955	2
0.1	1955.21	1955	3
-0.9	1955.29	1955	4
0.8	1955.38	1955	5
1.6	1955.46	1955	6
1.7	1955.54	1955	7
1.4	1955.63	1955	8
1.4	1955.71	1955	9
1.5	1955.79	1955	10
1.4	1955.88	1955	11

Example: Southern Oscillation

- Time plot

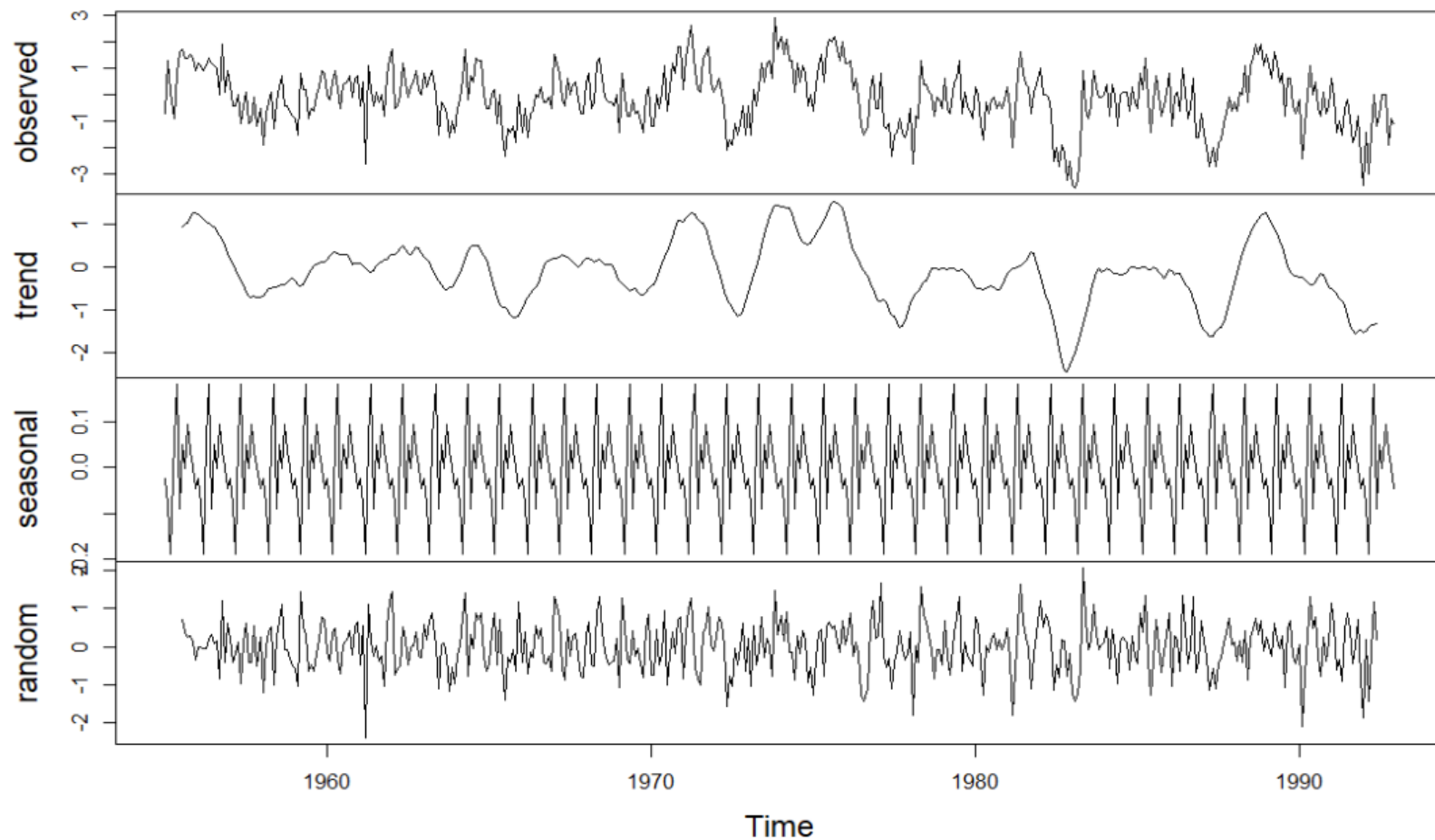


- No obvious periodic patterns

Example: Southern Oscillation

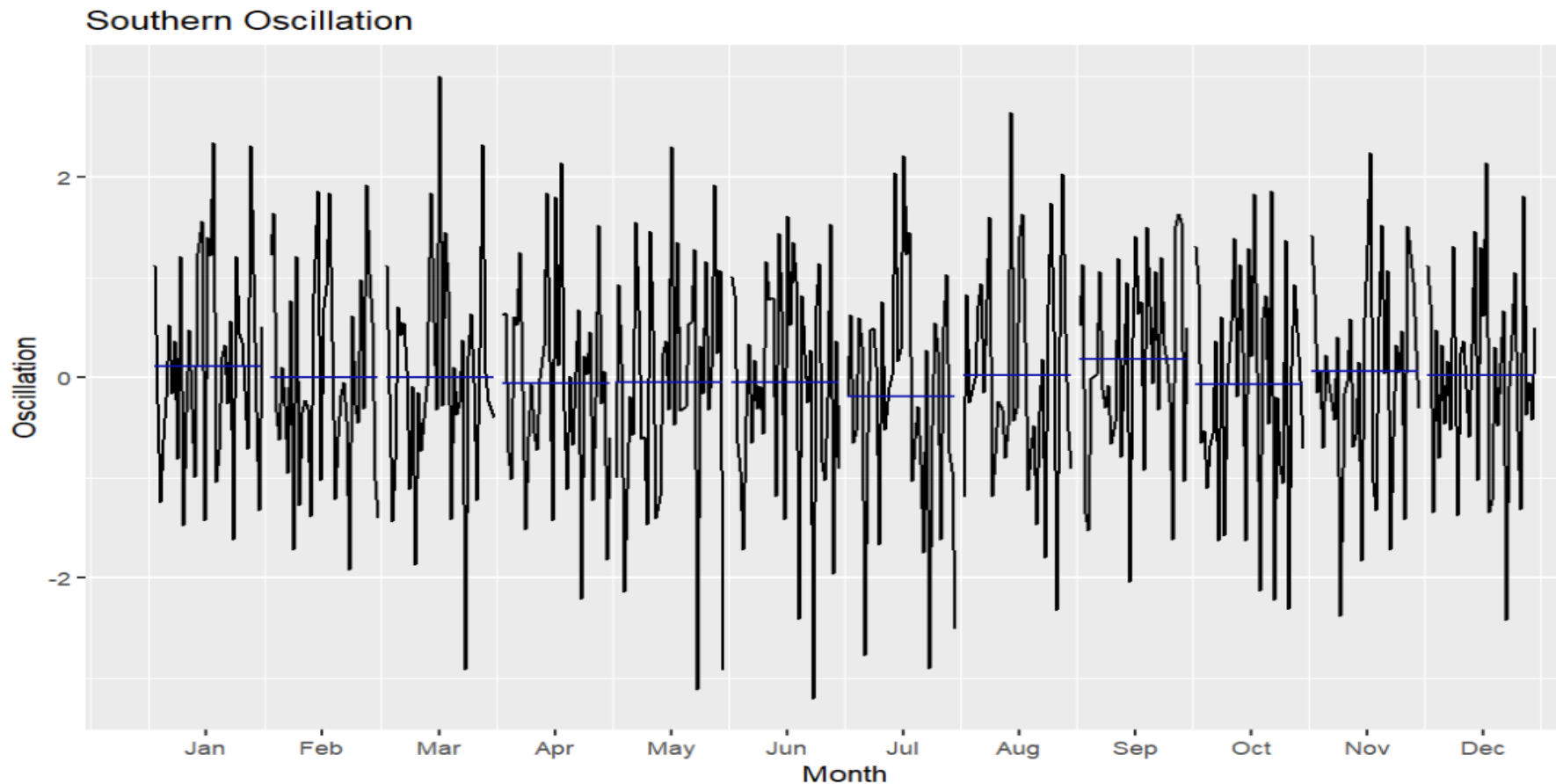
- Components

Decomposition of additive time series



Example: Southern Oscillation

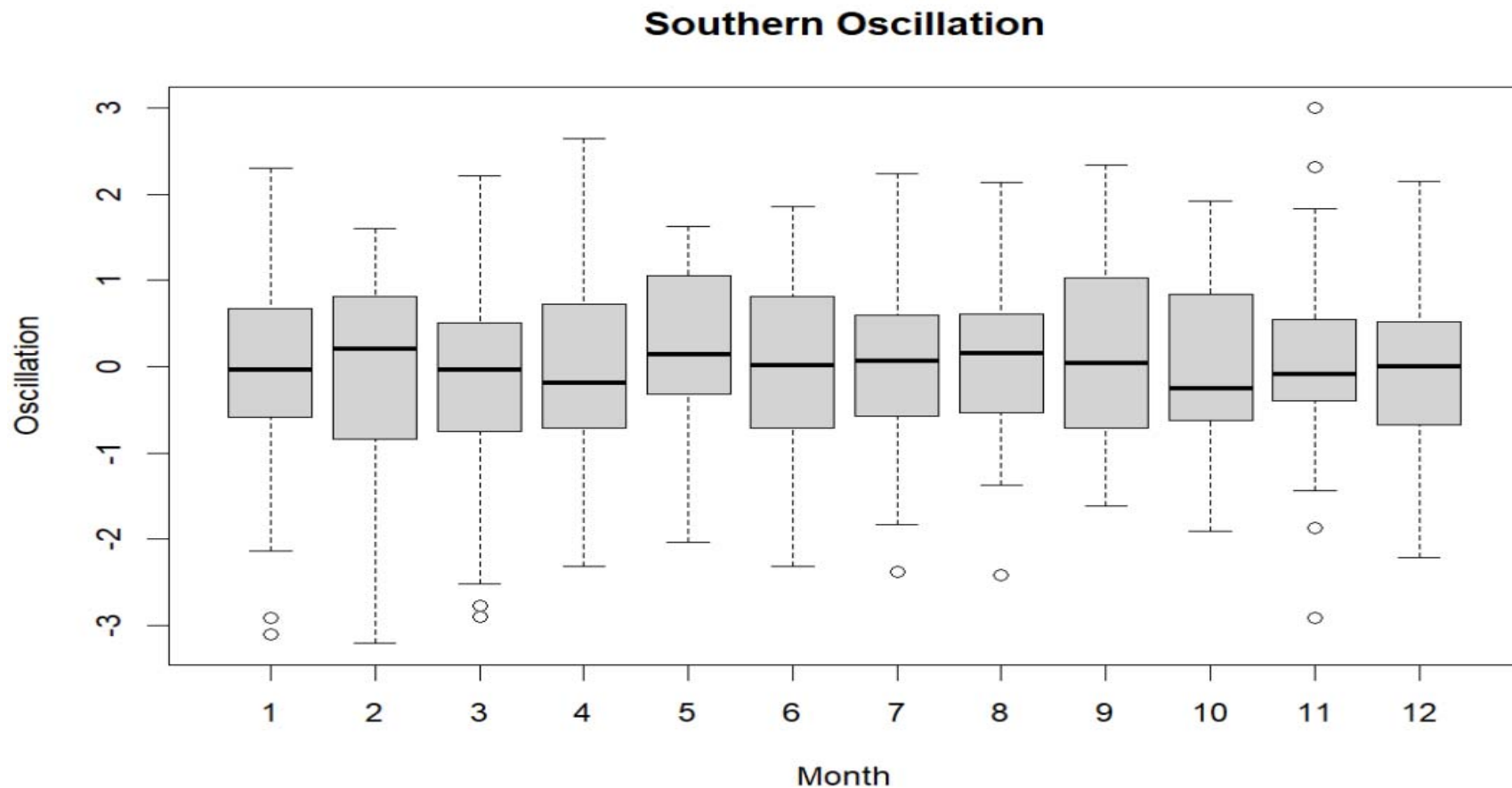
- Seasonal subseries plot



- No obvious seasonal pattern

Example: Southern Oscillation

- Boxplots



- No obvious seasonal pattern

Forecasting by Regression

- A regression model is assumed and parameters are estimated.
- The model can be used for future prediction.
- Can capture trend and/or seasonality.
- Trends:
 - Linear trend
 - Exponential trend
 - Polynomial trend

Forecasting by Regression

- Linear trend
 - Output variable Y is expressed as a linear function of input variable (or predictor variable) X .
 - Here, Y is the time series measurements and X is a time index.

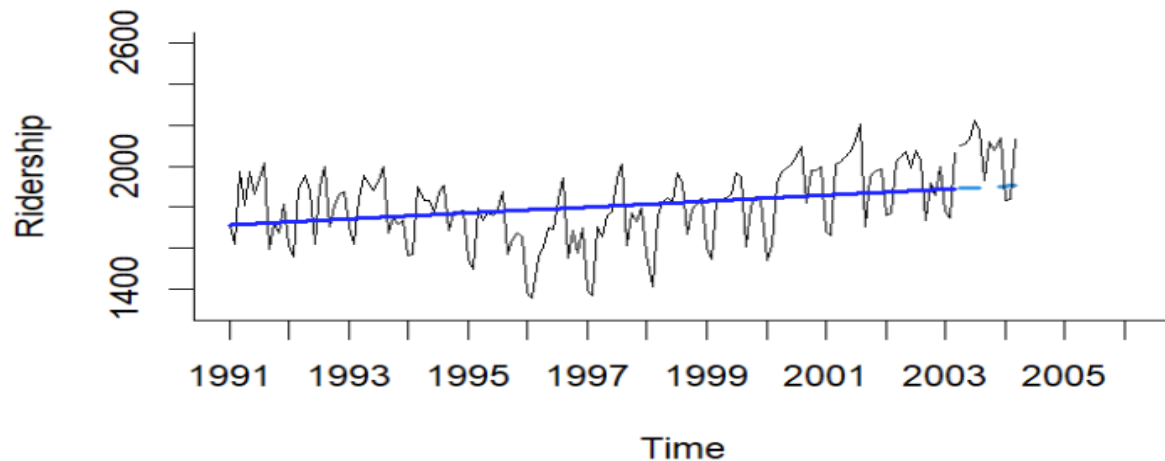
$$Y_t = b_0 + b_1 X_t + e, \text{ where}$$

b_0 and b_1 are regression coefficients

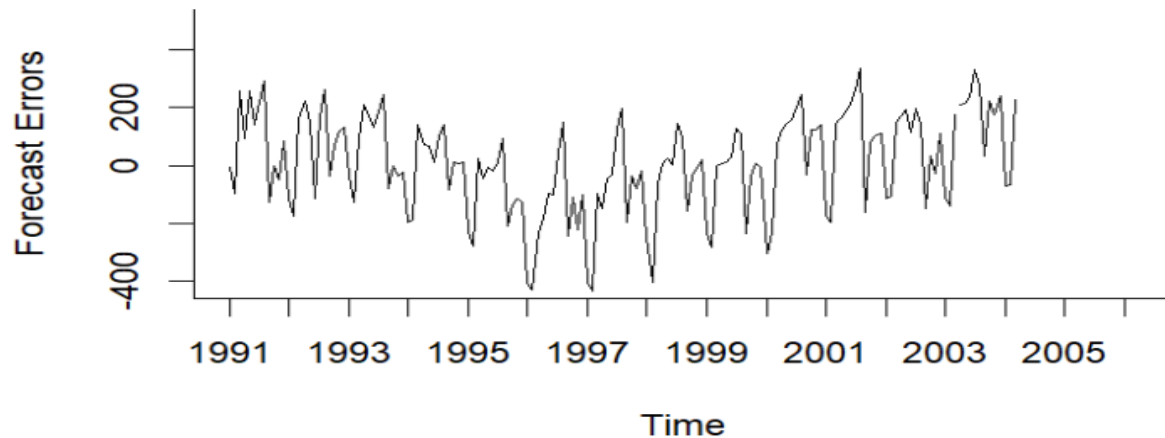
b_0 models level, b_1 models trend, e models noise

Forecasting by Regression

- Amtrak ridership (Jan 1991 – Mar 2004)

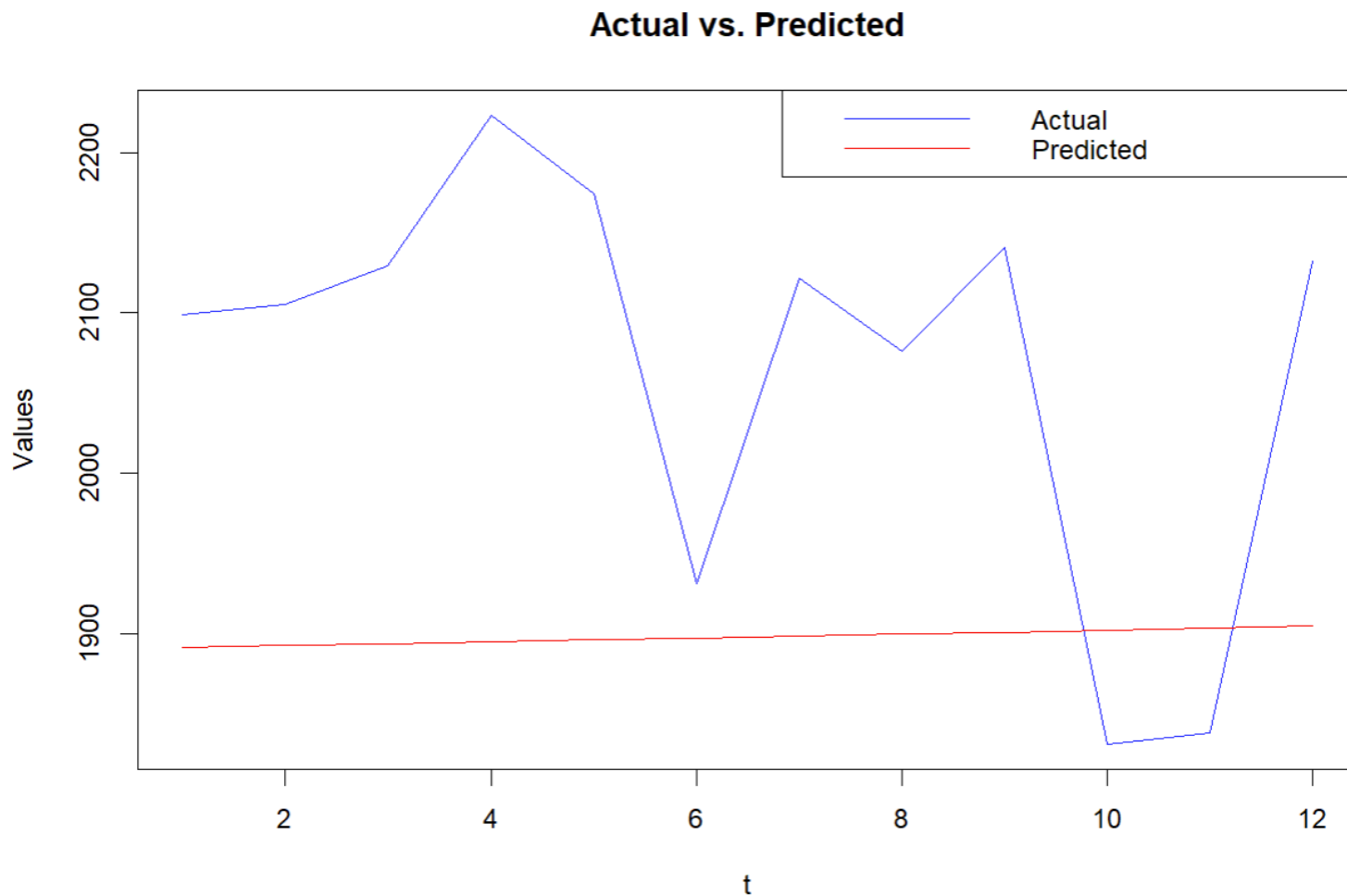


Coefficients
intercept (b_0): 1713.029
trend (b_1): 1.025



Forecasting by Regression

- Prediction on validation set



Forecasting by Regression

■ Prediction on validation set

Actual Ridership	Predicted Ridership	Error
2098.899	1891.415	207.484253
2104.911	1892.62	212.290942
2129.671	1893.825	235.845632
2223.349	1895.031	328.318321
2174.36	1896.236	278.12401
1931.406	1897.441	33.9646999
2121.47	1898.647	222.823389
2076.054	1899.852	176.202079
2140.677	1901.057	239.619768
1831.508	1902.263	-70.754543
1838.006	1903.468	-65.461853
2132.446	1904.673	227.772836

RMSE = 210.025

Forecasting by Regression

- Exponential trend
 - Captures multiplicative increase/decrease of time series values over time.

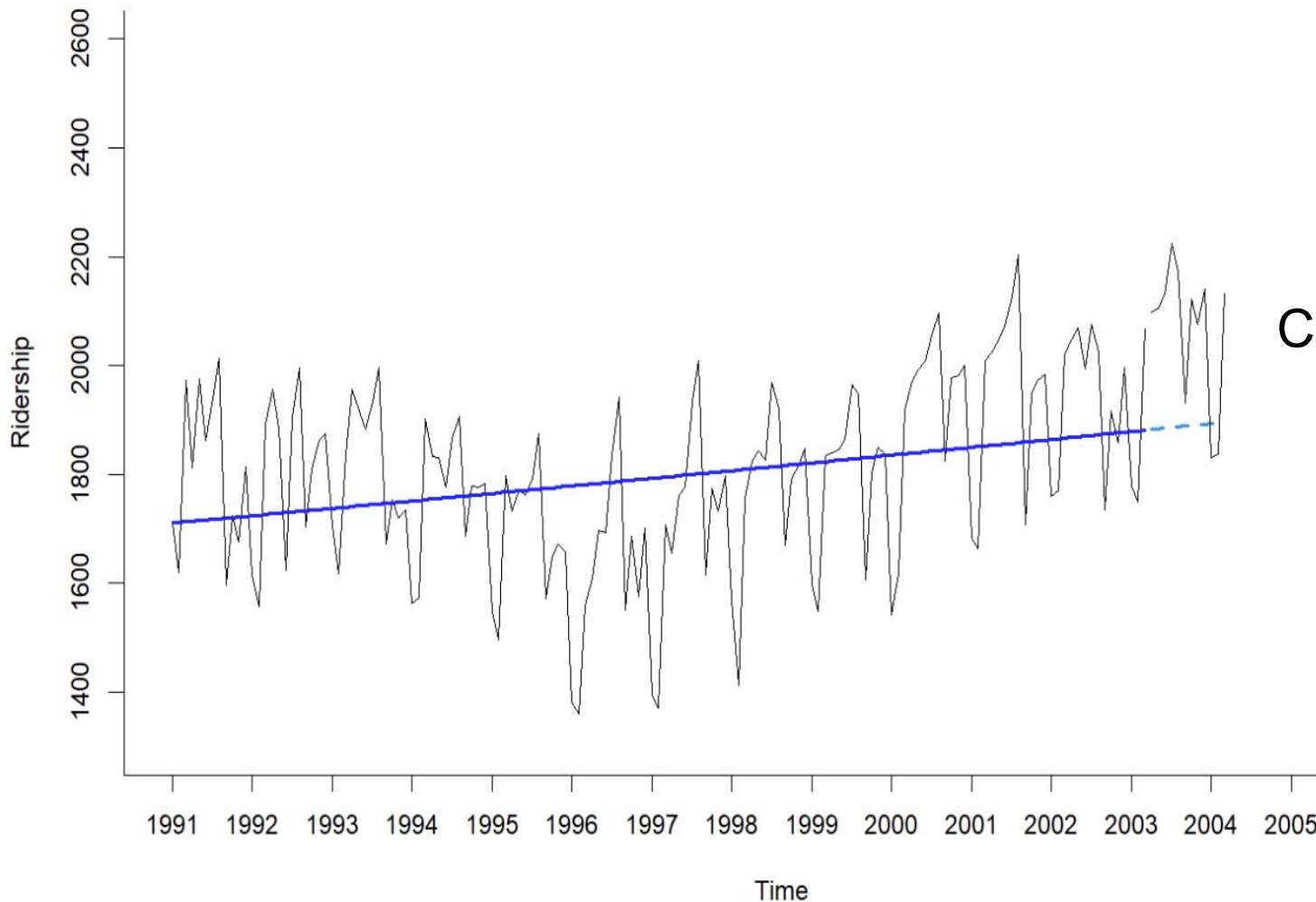
$$Y_t = ce^{b_1 X_t + e}$$

- Replace Y_t with $\log(Y_t)$ and fit a linear regression ($\log$ is the natural logarithm)

$$\log Y_t = b_0 + b_1 X_t + e$$

Forecasting by Regression

- Exponential trend



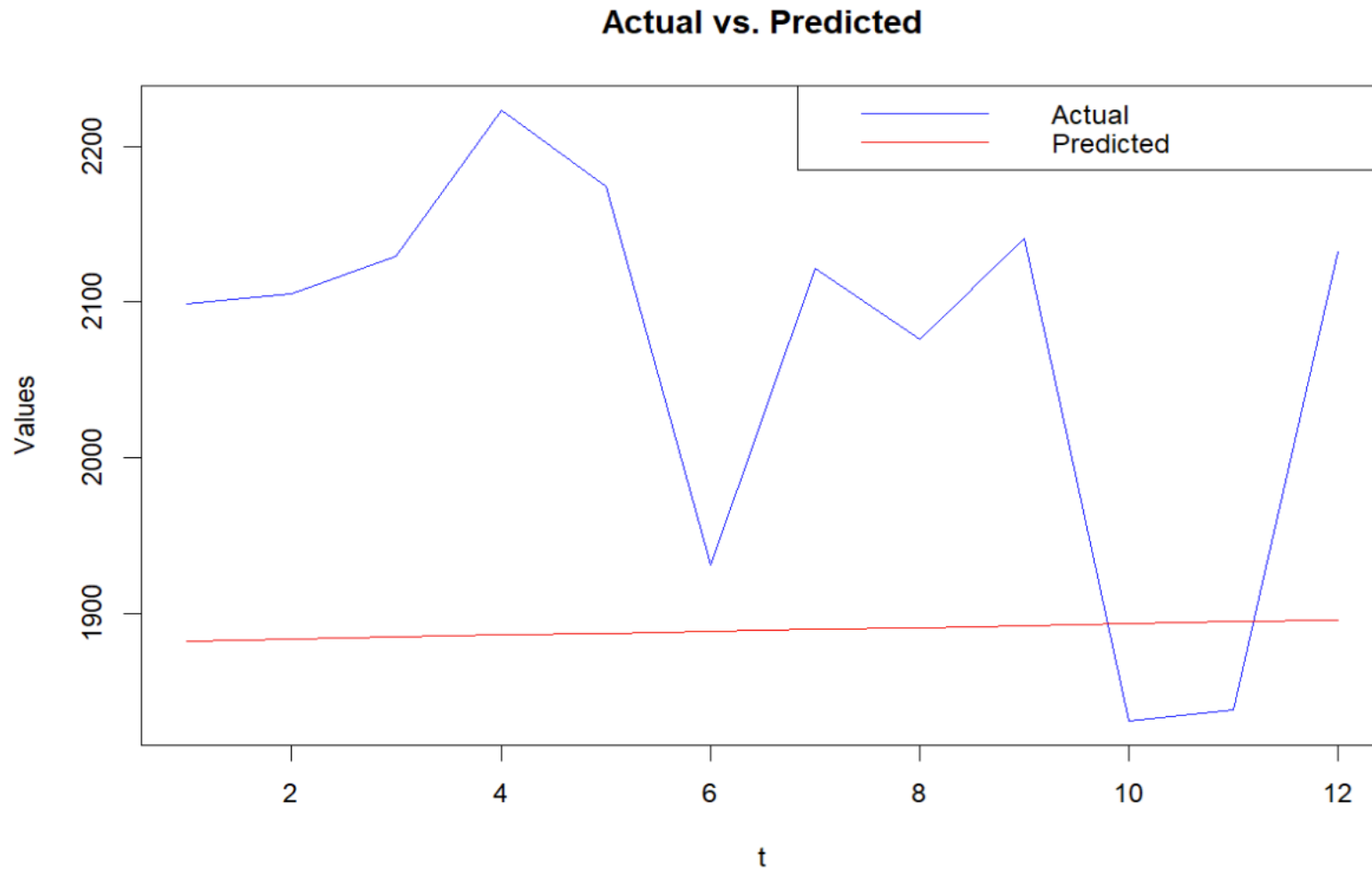
Coefficients:

Intercept (b_0) = 7.444,

Trend (b_1) = 0.00065

Forecasting by Regression

- Prediction on validation set



Forecasting by Regression

■ Prediction on validation set

Ridership	Log(ridership)	Predicted Log(ridership)	Predicted Ridership	Error
2098.89900	7.64917	7.54037	1882.52882	216.37018
2104.91100	7.65203	7.54102	1883.75521	221.15579
2129.67100	7.66372	7.54167	1884.98241	244.68859
2223.34900	7.70677	7.54233	1886.21040	337.13860
2174.36000	7.68449	7.54298	1887.43919	286.92081
1931.40600	7.56600	7.54363	1888.66878	42.73722
2121.47000	7.65986	7.54428	1889.89918	231.57082
2076.05400	7.63822	7.54493	1891.13037	184.92363
2140.67700	7.66888	7.54558	1892.36237	248.31463
1831.50800	7.51289	7.54623	1893.59517	-62.08717
1838.00600	7.51644	7.54688	1894.82877	-56.82277
2132.44600	7.66502	7.54754	1896.06318	236.38282

RMSE = 217.15

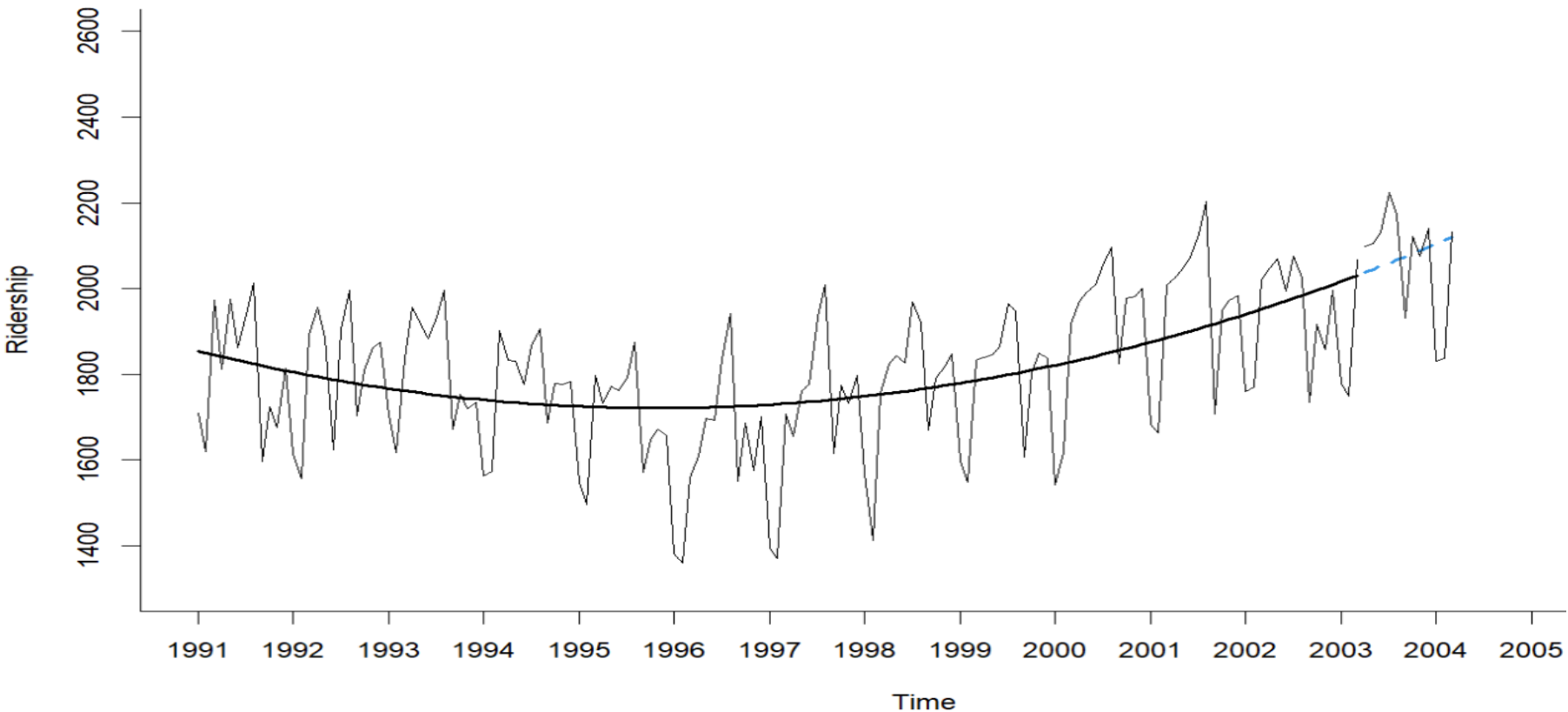
Forecasting by Regression

- Polynomial trend
 - Trend follows a polynomial shape.

$$Y_t = b_0 + b_1X_t + b_2X_t^2 + e$$

Forecasting by Regression

- Polynomial trend

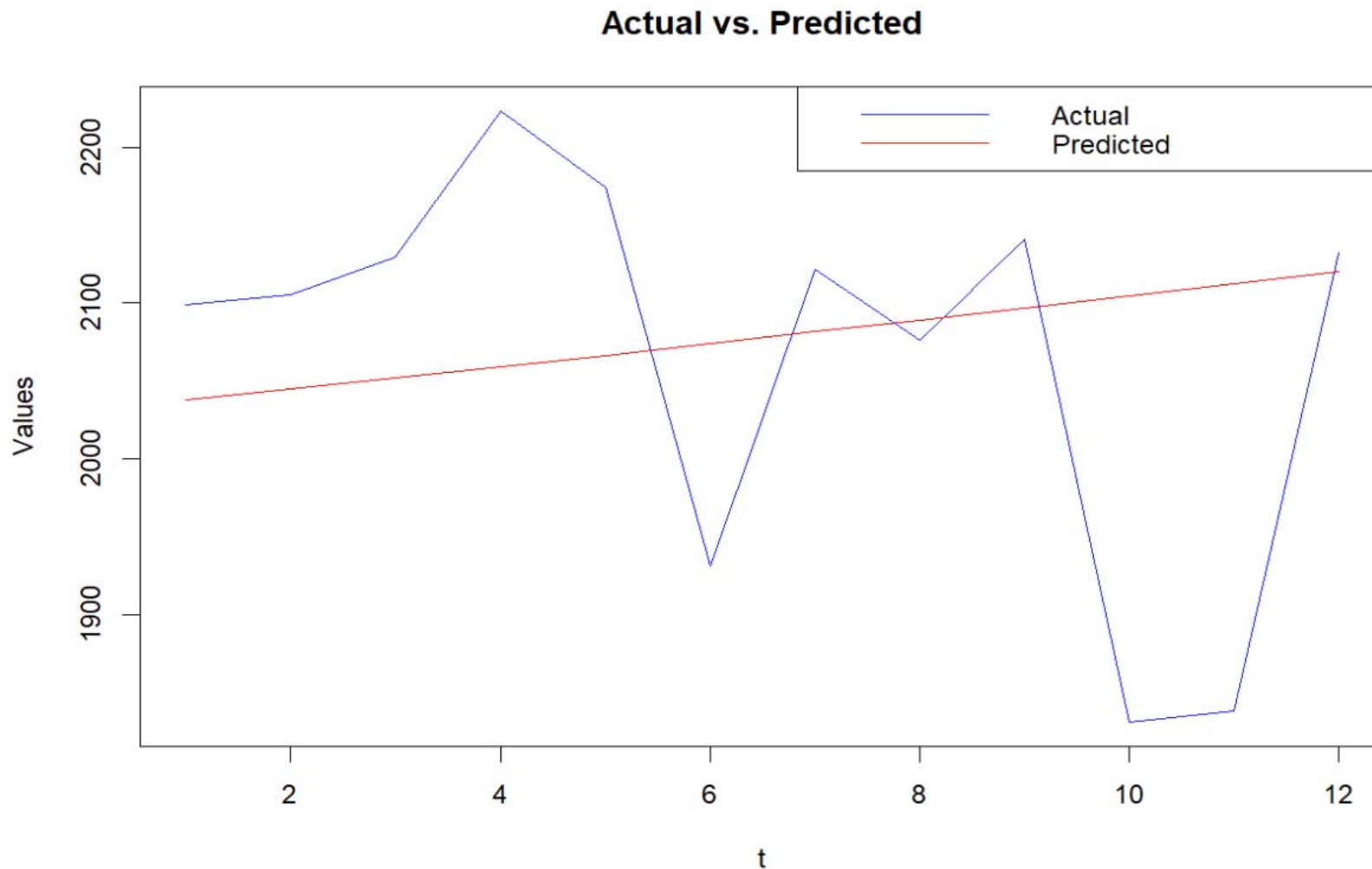


- Coefficients:

$$b_0 = 1859.1449, b_1 = -4.6785, b_2 = 0.0398$$

Forecasting by Regression

- Prediction on validation set



Forecasting by Regression

- Prediction on validation set

Actual Ridership	Predicted Ridership	Error
2098.899	2037.530893	61.36811
2104.911	2044.659831	60.25117
2129.671	2051.868281	77.80272
2223.349	2059.156242	164.1928
2174.36	2066.523716	107.8363
1931.406	2073.970701	-142.565
2121.47	2081.497198	39.9728
2076.054	2089.103207	-13.0492
2140.677	2096.788727	43.88827
1831.508	2104.55376	-273.046
1838.006	2112.398304	-274.392
2132.446	2120.32236	12.12364

RMSE = 137.25

Forecasting by Regression

- Model with seasonality
 - Create a new categorical variable *season*, whose values are *Jan, Feb, Mar, ...*
 - Then, convert the *season* variable to 11 indicator (or dummy) variables.
 - This is done by R automatically.
 - Coefficients: intercept and 11 indicator variables.

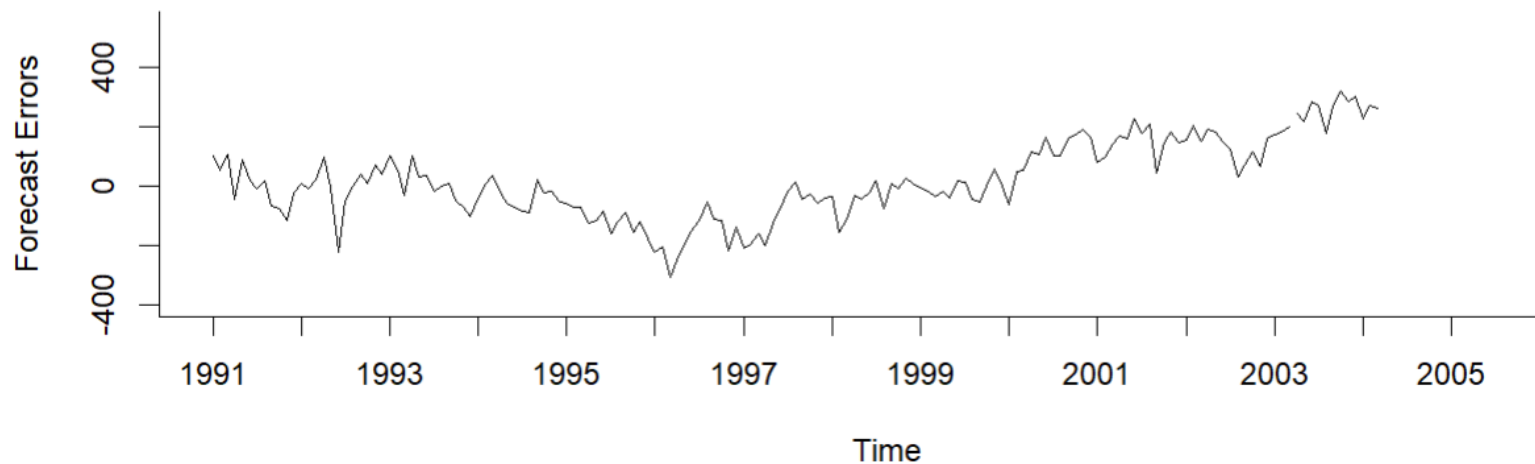
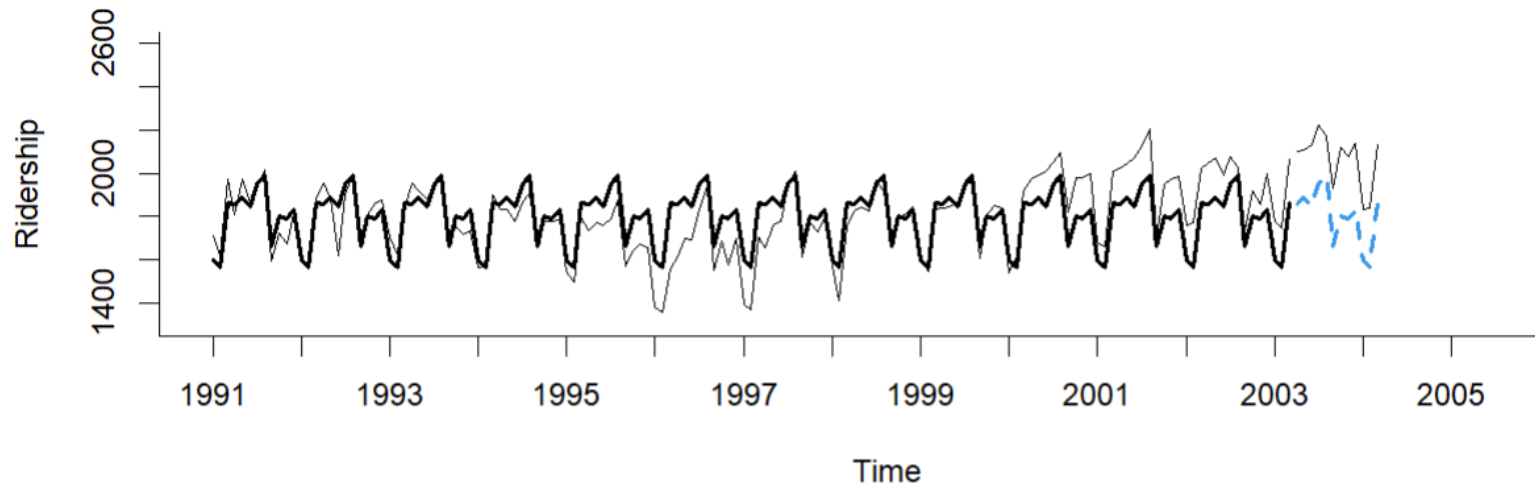
Forecasting by Regression

- Model with seasonality
 - Estimated coefficients

```
Coefficients:
              Estimate
(Intercept)  1603.95
season2      -37.68
season3      262.86
season4      251.29
season5      282.53
season6      241.17
season7      345.63
season8      390.68
season9       57.65
season10     197.16
season11     187.32
season12     231.46
```

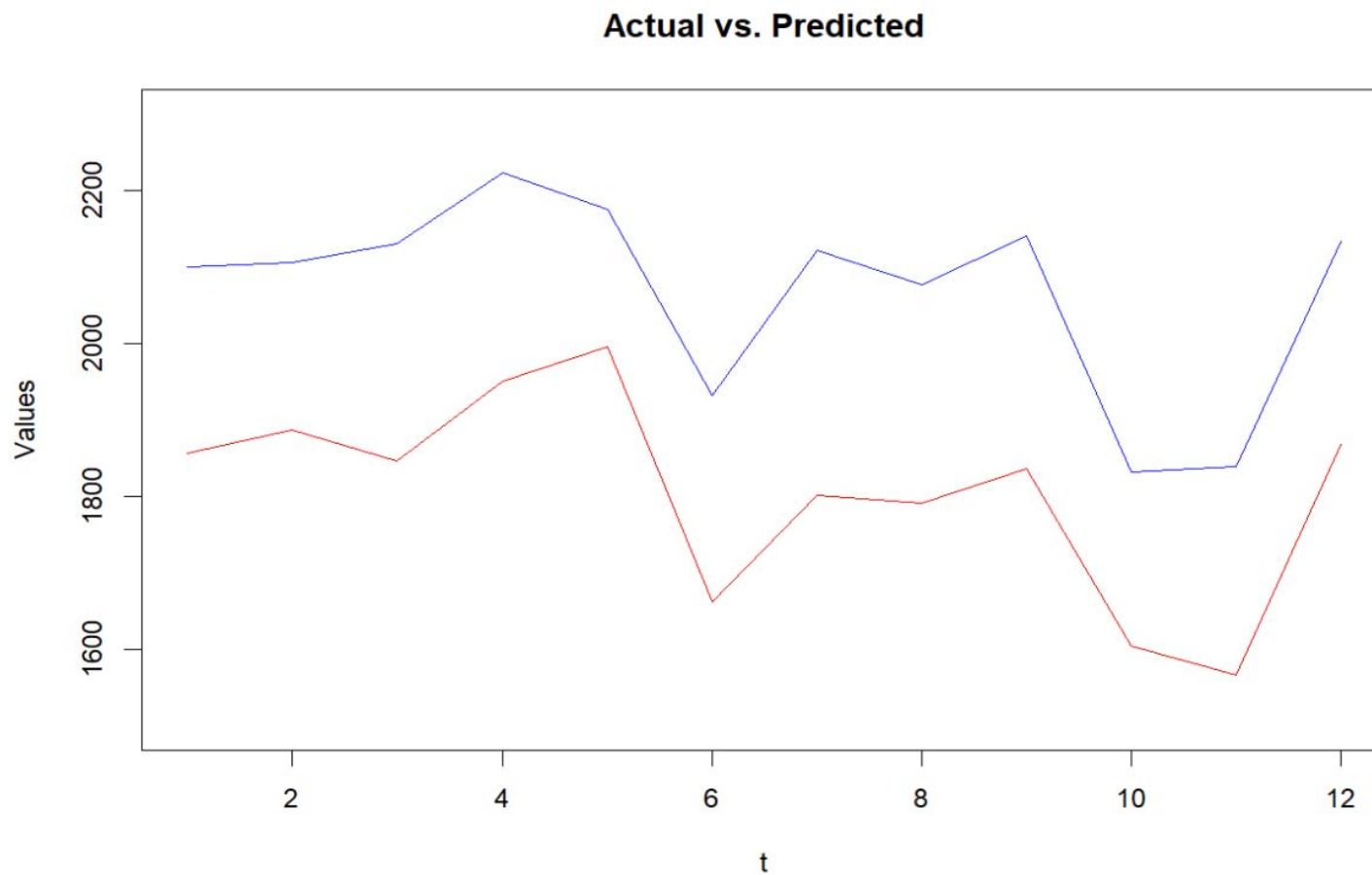
Forecasting by Regression

■ Plot



Forecasting by Regression

- Prediction on validation set



Forecasting by Regression

- Prediction on validation set

Ridership	Predicted	Error
2098.899	1855.236	243.6631
2104.911	1886.476	218.4348
2129.671	1845.125	284.546
2223.349	1949.58	273.7688
2174.36	1994.626	179.7338
1931.406	1661.599	269.8073
2121.47	1801.107	320.3629
2076.054	1791.269	284.7846
2140.677	1835.413	305.2642
1831.508	1603.95	227.5575
1838.006	1566.273	271.7332
2132.446	1866.809	265.637

RMSE = 264.77

Forecasting by Regression

- Model with trend and seasonality

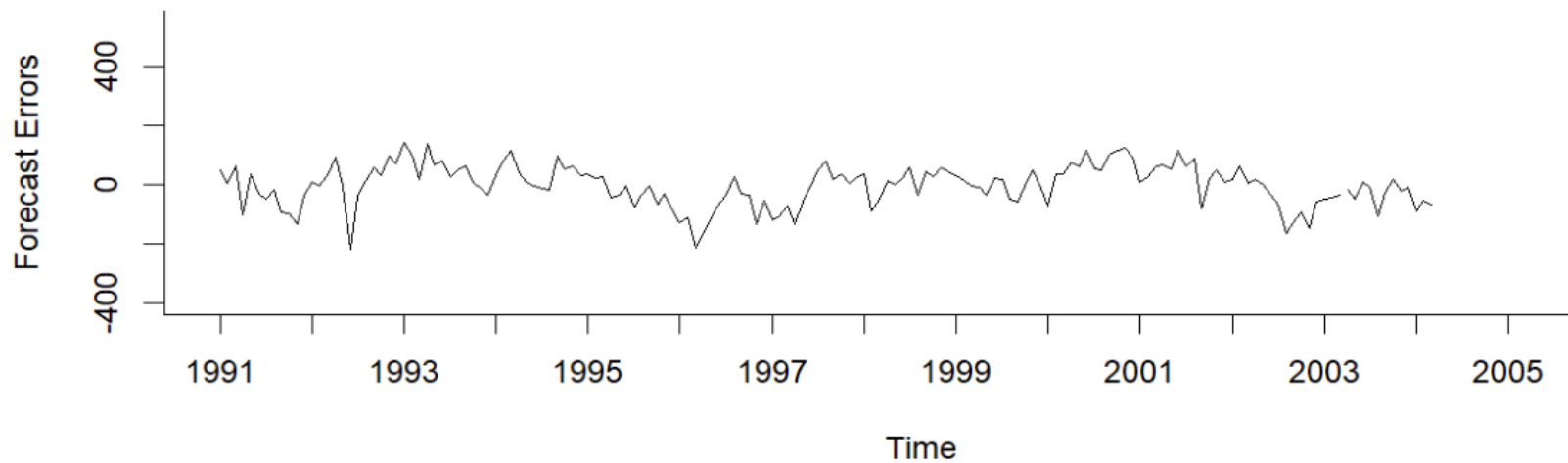
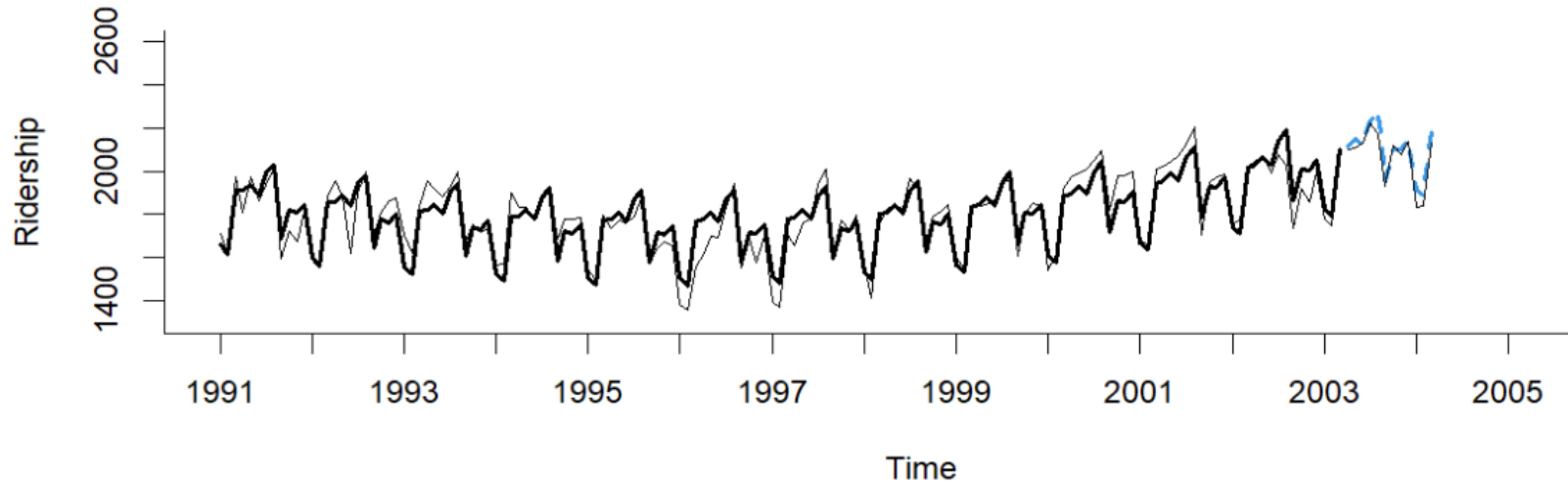
- A model captures both trend and seasonality.
- It appears that Amtrak time series has quadratic trend and seasonality.
- Fit model with 13 predictors:
 t and t^2 for quadratic trend
11 season indicator variables

Coefficients:

	Estimate
(Intercept)	1.666e+03
trend	-5.247e+00
I(trend^2)	4.376e-02
season2	-3.886e+01
season3	2.604e+02
season4	2.674e+02
season5	2.978e+02
season6	2.554e+02
season7	3.588e+02
season8	4.026e+02
season9	6.832e+01
season10	2.065e+02
season11	1.952e+02
season12	2.378e+02

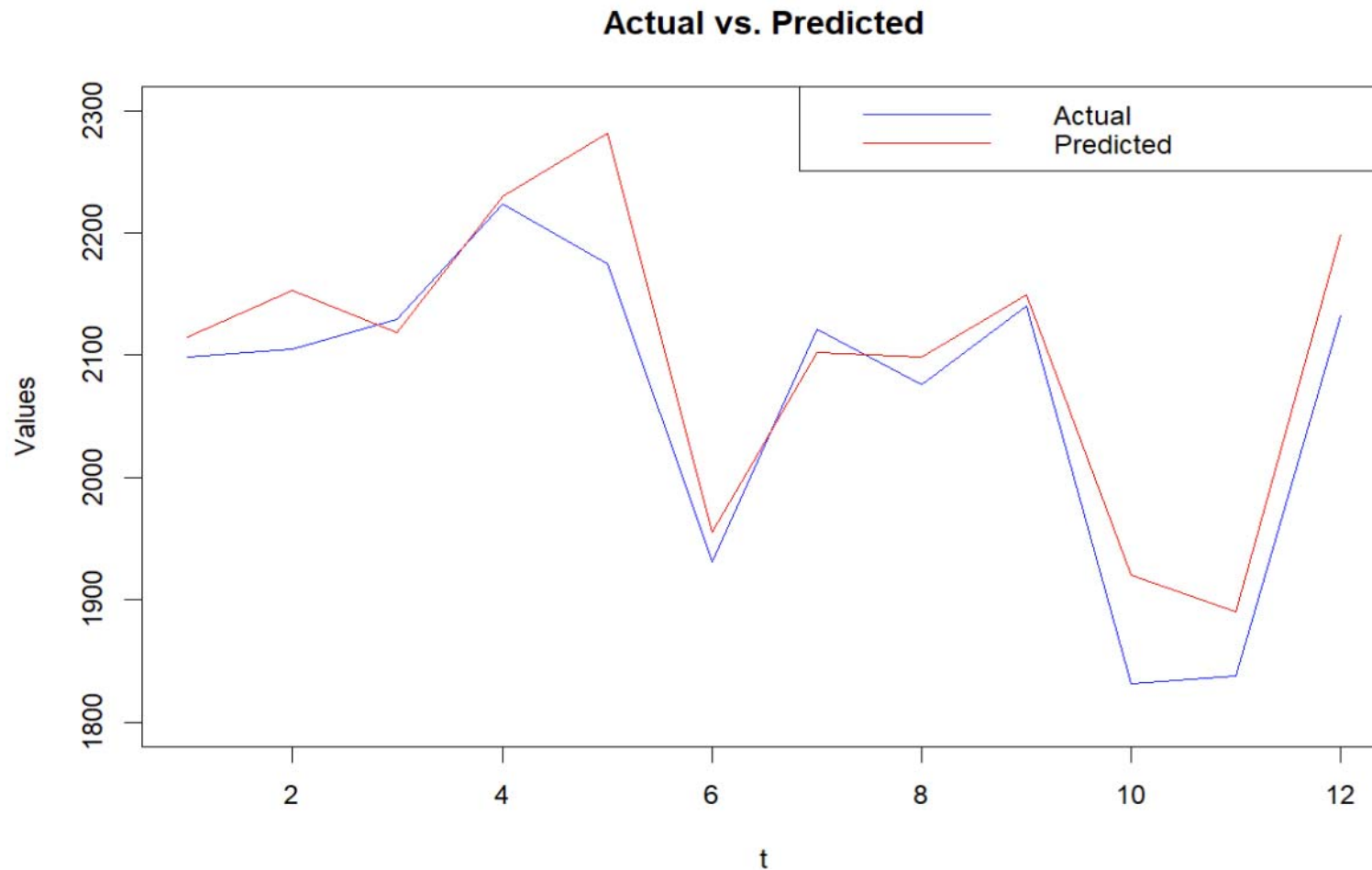
Forecasting by Regression

■ Plot



Forecasting by Regression

- Prediction on validation set



Forecasting by Regression

- Prediction on validation set

Ridership	Predicted	Error
2098.899	2114.958	-16.0592
2104.911	2153.025	-48.1135
2129.671	2118.499	11.17168
2223.349	2229.781	-6.43151
2174.36	2281.653	-107.293
1931.406	1955.451	-24.0452
2121.47	2101.786	19.68448
2076.054	2098.774	-22.7199
2140.677	2149.743	-9.06633
1831.508	1920.407	-88.8989
1838.006	1890.08	-52.0743
2132.446	2197.968	-65.5217

RMSE = 50.599

Smoothing Methods

- Data driven (i.e., not based on a theoretical model such as a regression model)
- Data taken over time contains some form of random variations
- Smoothing reduces or cancels the effect due to random variations
- Smoothing can reveal more clearly the underlying trend, seasonal and cyclic components.
- Will discuss moving average and exponential smoothing

Moving Average

- Centered moving average for visualization

Centered moving average with window = 3

$$M_t = (M_{t+1} + M_t + M_{t-1})/3$$

Year	\$ (millions)	CMA3
1985	46.16	
1986	47.00	46.99233
1987	47.82	47.70833
1988	48.31	48.295
1989	48.76	48.74433
1990	49.16	49.15667
1991	49.55	49.209
1992	48.92	49.59267
1993	50.32	49.99933
1994	50.77	

Moving Average

- Centered moving average is good for visualizing trends because it suppresses seasonality and noise.
- Not for forecasting because future values are not known.

Moving Average

- Trailing moving average for forecasting

Moving average with window = 3

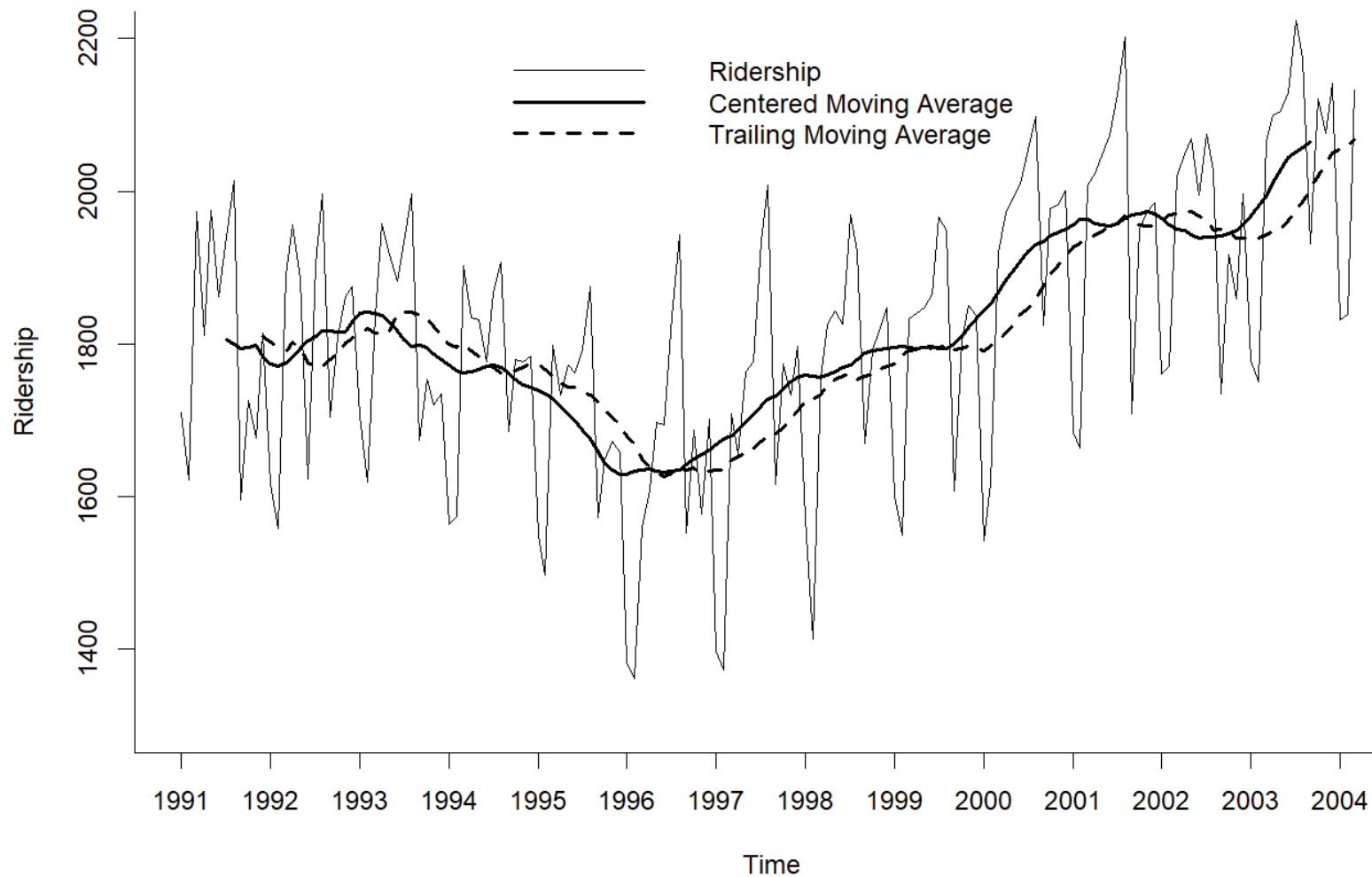
$$M_{t+1} = (M_t + M_{t-1} + M_{t-2})/3$$

Year	\$ (millions)	MA3
1985	46.16	
1986	47.00	
1987	47.82	
1988	48.31	46.99333
1989	48.76	47.71
1990	49.16	48.29667
1991	49.55	48.74333
1992	48.92	49.15667
1993	50.32	49.21
1994	50.77	49.59667

RMSE = 0.992

Moving Average

- Centered and trailing moving average (Amtrak)



Moving Average

- Window size
 - Large window: reveals more global trends
 - Small window: reveals local trends
 - Better to empirically decide by experimenting with different window sizes
- Plot in previous slide shows moving average is not accurate for Amtrak data.
- Moving average can be used only for time series without trend and seasonality.

Exponential Smoothing

- In moving average, observations in the past used in calculating the average are weighted equally.
- Exponential Smoothing assigns *exponentially decreasing weights* as the observations get older.
- In other words, *recent observations are given relatively more weight in forecasting than older observations.*

Exponential Smoothing

- Single exponential smoothing

$$F_2 = X_1$$

$$F_t = \alpha X_{t-1} + \alpha (1 - \alpha) X_{t-2} + \alpha (1 - \alpha)^2 X_{t-3} + \dots, \text{ for } t \geq 3$$

$$0 < \alpha \leq 1$$

F_t can also be calculated as follows, which is easier in practice:

$$F_2 = X_1$$

$$F_t = \alpha X_{t-1} + (1 - \alpha) F_{t-1}, \text{ for } t \geq 3$$

Exponential Smoothing

- Single exponential smoothing

Time	Xt	F($\alpha=0.1$)	F($\alpha=0.5$)
1	71		
2	70	71	71
3	69	70.9	70.5
4	68	70.71	69.75
5	64	70.44	68.88
6	65	69.8	66.44
7	72	69.32	65.72
8	78	69.58	68.86
9	75	70.43	73.43
10	75	70.88	74.21
11	75	71.29	74.61
12	70	71.67	74.80

$\alpha=0.1$

RMSE = 4.359

$\alpha=0.5$

RMSE = 4.036

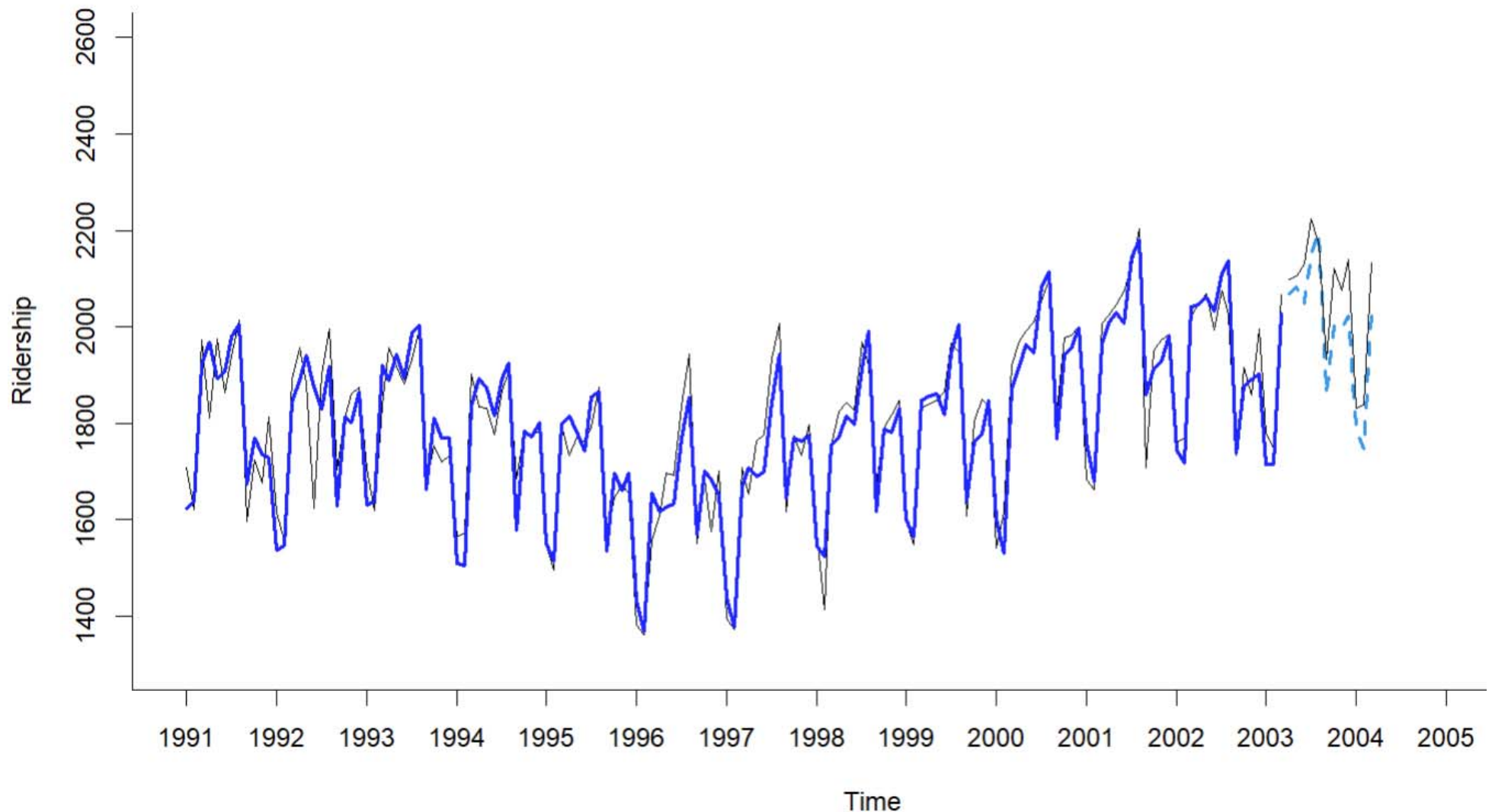
$\alpha=0.5$ is better

Exponential Smoothing

- Single exponential smoothing does not do well when the time series data has a trend.
- Double exponential smoothing is better in handling trends.
- If the data has both trend and seasonality, double exponential smoothing does not work.
- Triple exponential smoothing (Holt-Winter's method) works well with time series with trend and seasonality.

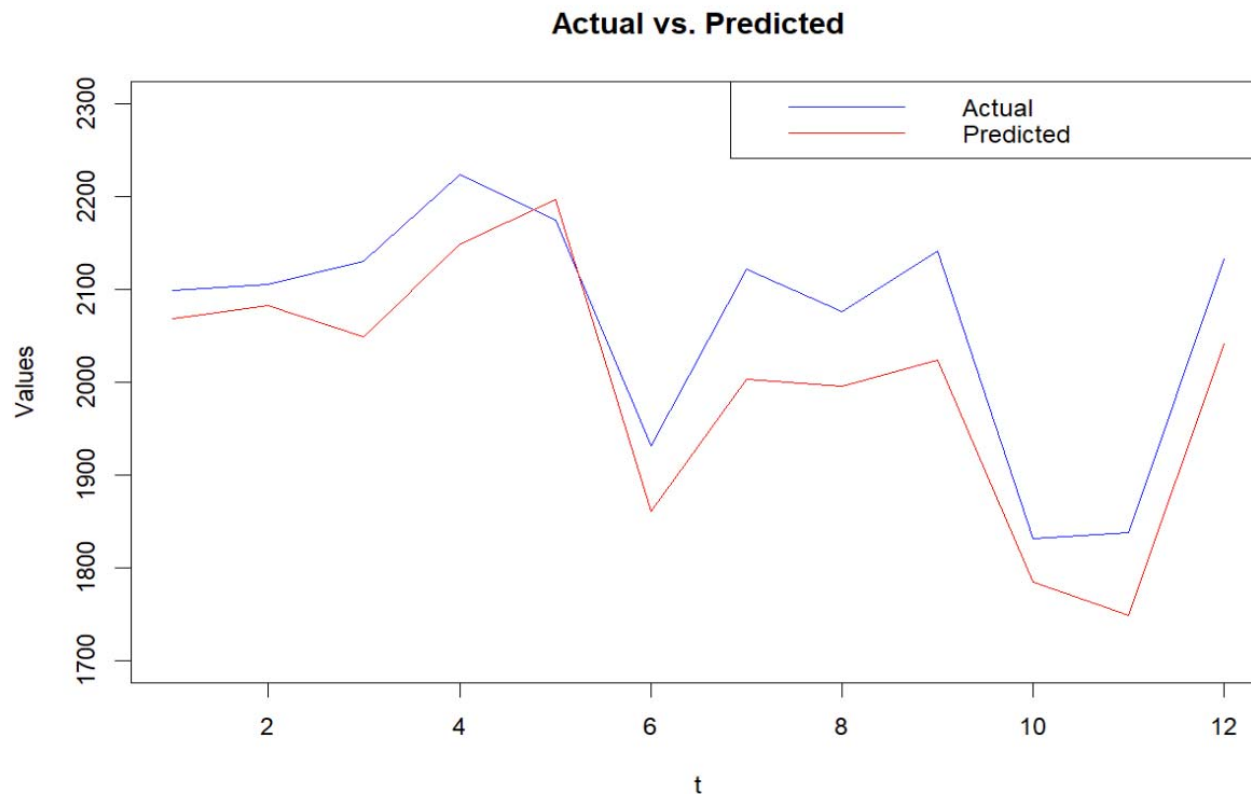
Exponential Smoothing

- Holt-Winter triple exponential smoothing



Exponential Smoothing

- Holt-Winter triple exponential smoothing
- Fit Holt-Winter model
ets(train.ts, model = "MAA")
- RMSE on test set = 77.427



FYI

- Holt double exponential smoothing

$$F_{t+k} = L_t + kT_t$$

$$L_t = \alpha X_t + (1 - \alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta) T_{t-1}$$

- Note:
 - L_t : Level estimate at time t
 - T_t : Trend estimate at time t

FYI

- Holt-Winter's triple exponential smoothing

$$F_{t+k} = (L_t + kT_t)S_{t+k-M} \quad // \text{ multiplicative seasonal}$$

$$L_t = \alpha \left(\frac{X_t}{S_{t-M}} \right) + (1 - \alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$$

$$S_t = \gamma \left(\frac{X_t}{L_t} \right) + (1 - \gamma)S_{t-M}$$

- Note:
 - S_t : Seasonality estimate at time t
 - M : Number of seasons

Autocorrelation

- Autocorrelation captures the dependence between time series values.
- Autocorrelation is a correlation between a time series and *lagged* version of the time series.
- Strong autocorrelation (positive or negative) at a lag k (>1) and its multiples ($2k$, $3k$, . . .) indicates there is a seasonality (a cyclical pattern).
- Positive lag-1 autocorrelation indicates that consecutive values move in the same direction, in general.

Autocorrelation

Month	Ridership	Lag-1	Lag-2
Jan-91	1709		
Feb-91	1621	1709	
Mar-91	1973	1621	1709
Apr-91	1812	1973	1621
May-91	1975	1812	1973
Jun-91	1862	1975	1812
Jul-91	1940	1862	1975
Aug-91	2013	1940	1862
Sep-91	1596	2013	1940
Oct-91	1725	1596	2013
Nov-91	1676	1725	1596
Dec-91	1814	1676	1725

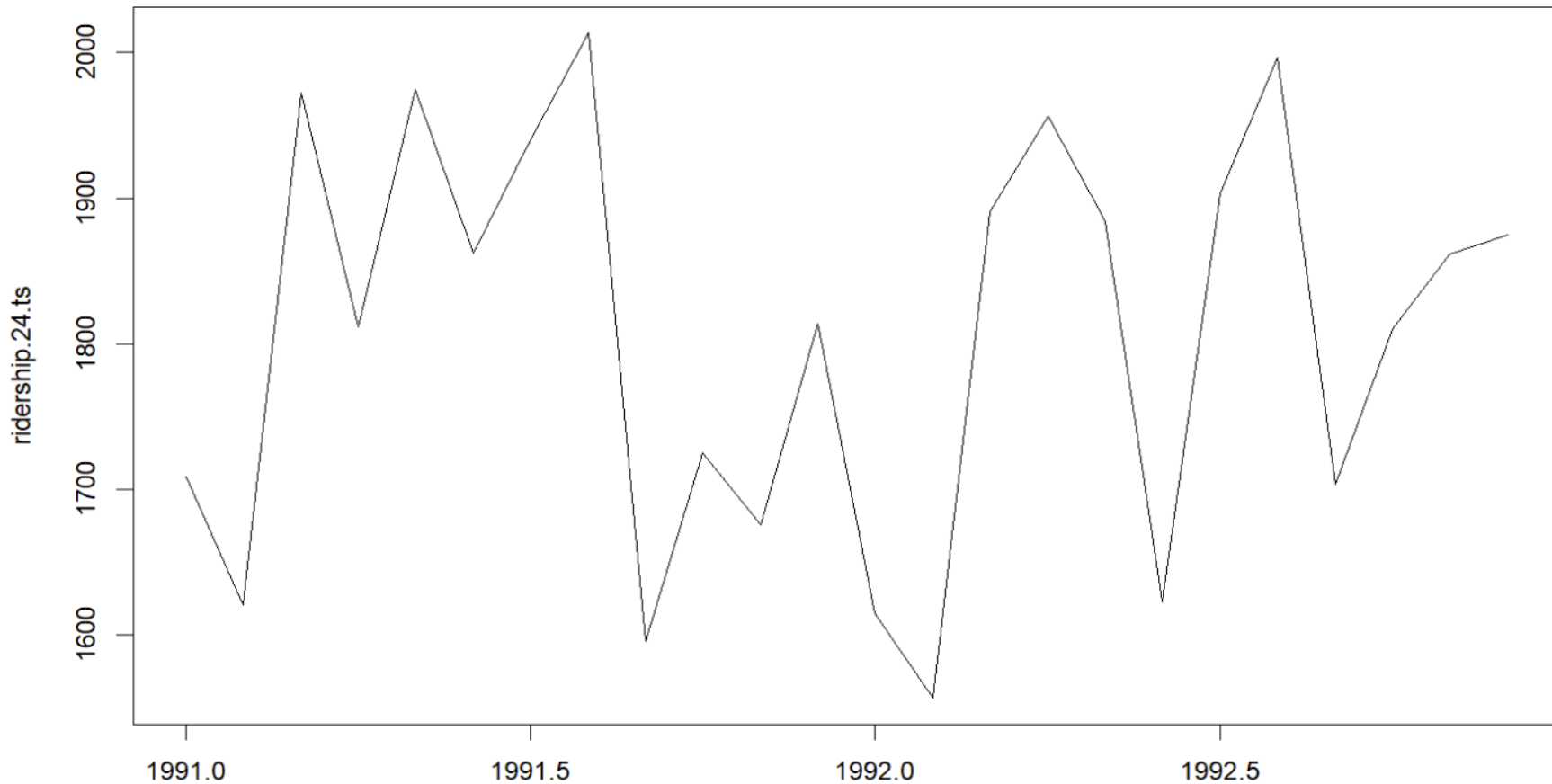
Lag-1 autocorrelation uses 11 values (Feb-91 through Dec-91)

Lag-2 autocorrelation uses 10 values (Mar-91 through Dec-91)

Lag-1 autocorrelation	0.072287
Lag-2 autocorrelation	0.113293

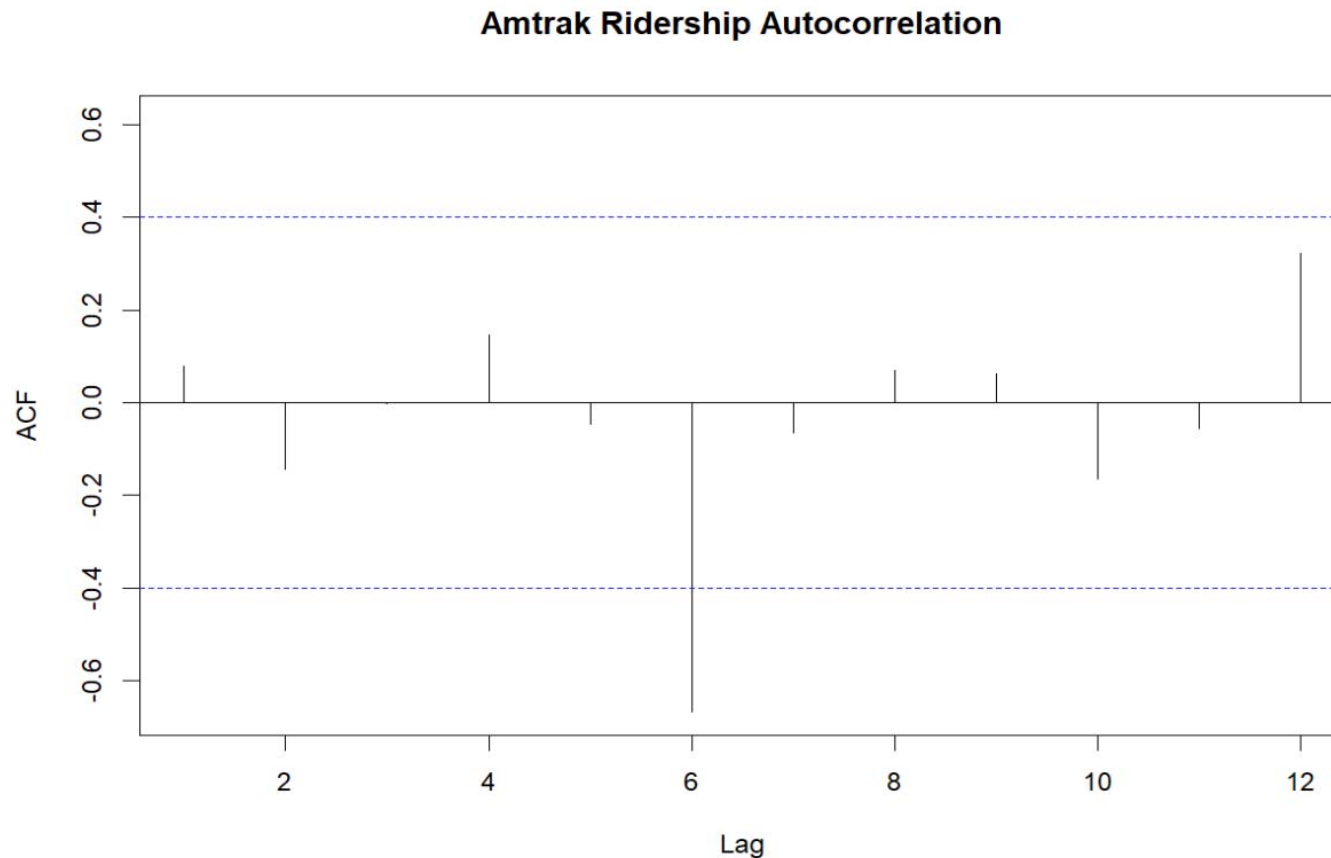
Autocorrelation

- Amtrak data example (first 24 months shown)



Autocorrelation

- Amtrak data example (for first 24 months)



- Strong correlation at lag-6.
- Indicates bi-annual seasonality; 6-month swing between low and high

Autocorrelation

- Autocorrelation plots can be also used for checking randomness in a data set.
- This randomness is ascertained by computing autocorrelations for data values at varying time lags.
- If random, such autocorrelations should be near zero for any and all time-lag separations.
- If non-random, then one or more of the autocorrelations will be significantly non-zero.

Autoregression (AR) Models

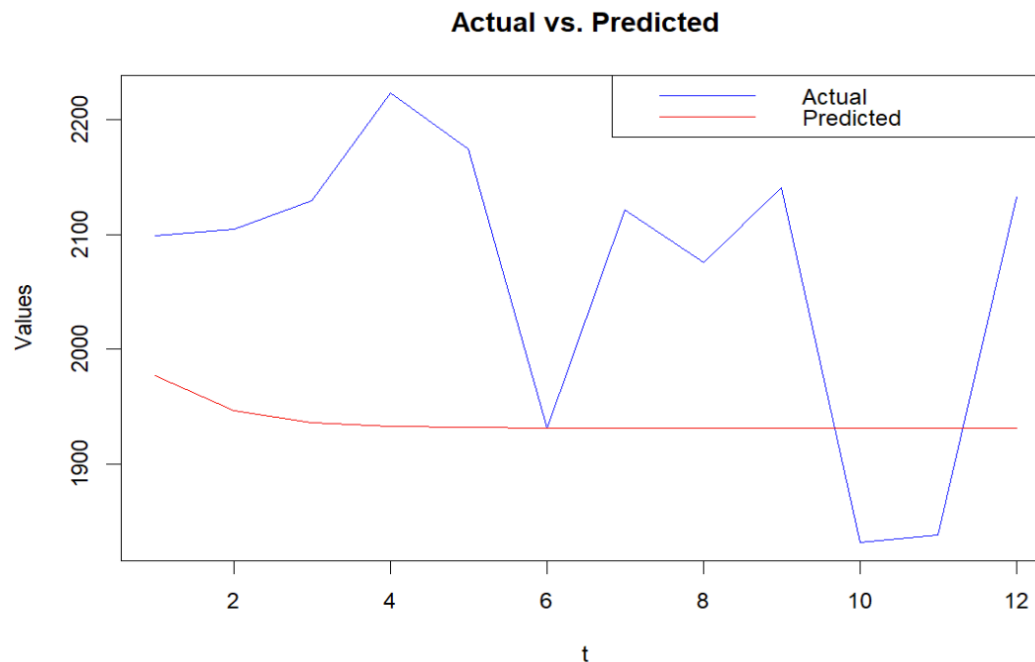
- Incorporate autocorrelation
- Similar to linear regression models but past time series values are used as predictors
- Example: AR(2), an autoregression model of order 2:

$$Y_t = b_0 + b_1 Y_{t-1} + b_2 Y_{t-2} + e$$

- ARIMA (autoregressive integrated moving average)
 - Combines autoregression and moving average.
 - More flexible but requires statistical expertise.

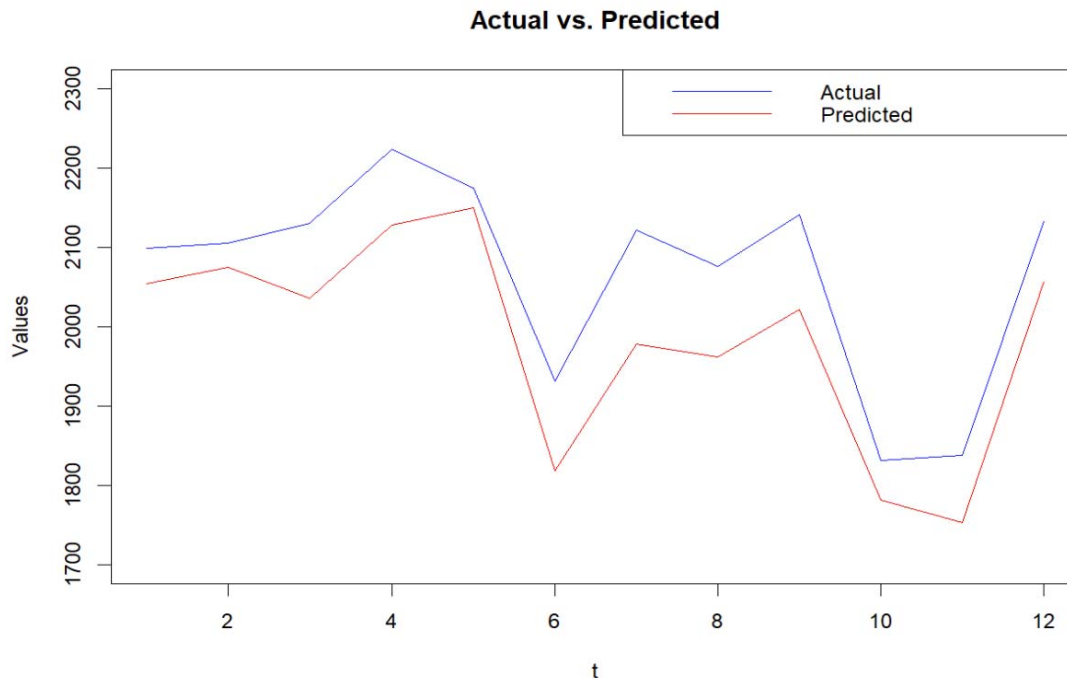
ARIMA

- Examples using Amtrak data
- Fit arima model without seasonality, $p = 1$, $d = 1$, $q = 1$
`arima(train.ts, order=c(1,1,1), method="ML")`
- RMSE on validation set = 178.219



ARIMA

- Fit arima model with seasonality
non-seasonal: $p = 1, d = 1, q = 1$
seasonal: $P = 1, D = 1, Q = 1$
`arima(train.ts, order=c(1,1,1), seasonal = list(order = c(1,1,1), period = 12), method="ML")`
- RMSE on validation set = 90.420



ARIMA

- Use *auto.arima* to find parameters

```
auto.arima.fit <- auto.arima(train.ts)  
arimaorder(auto.arima.fit)
```

- Parameters found by *auto.arima*

<i>p</i>	<i>d</i>	<i>q</i>	<i>P</i>	<i>D</i>	<i>Q</i>	<i>Frequency</i>
1	1	1	0	1	1	12

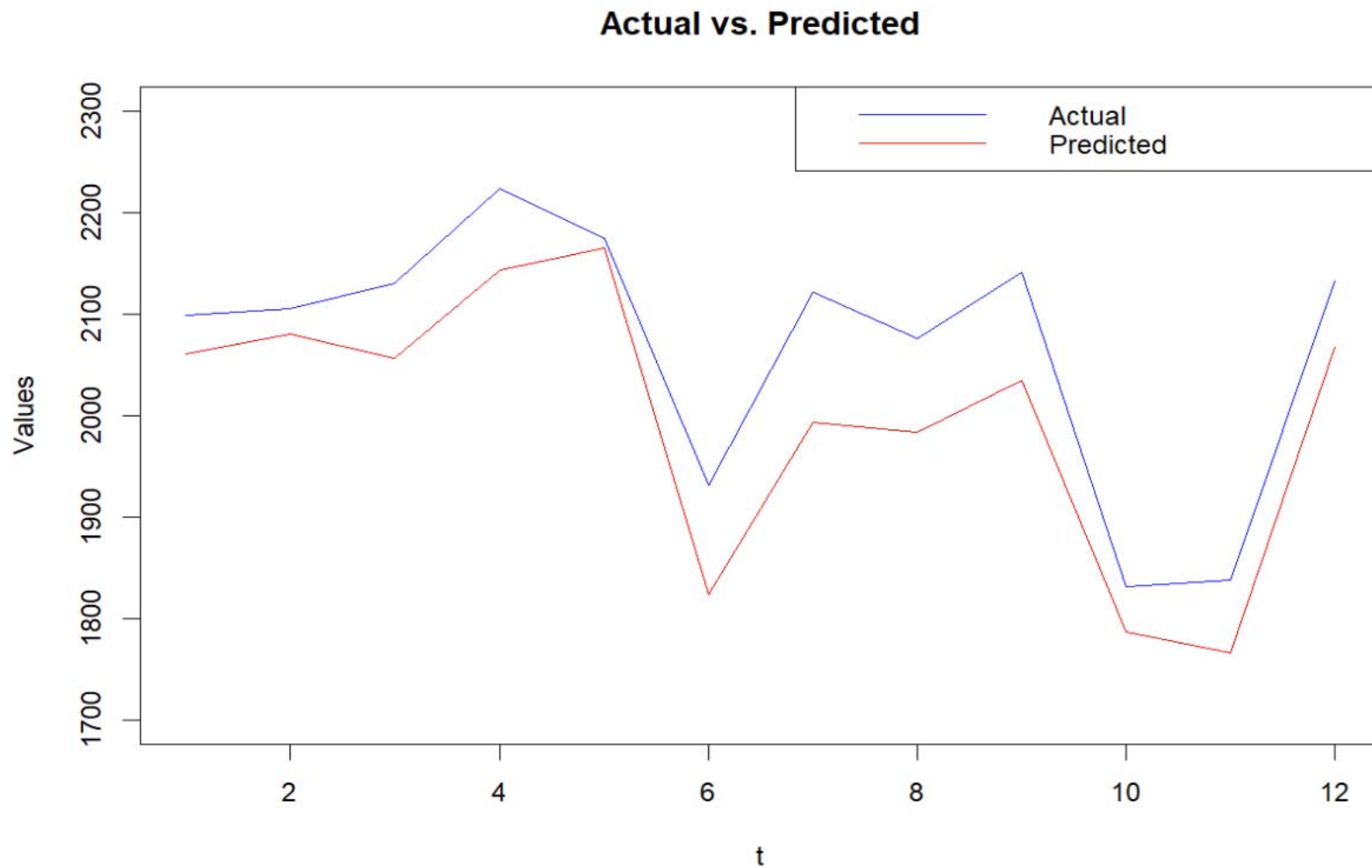
non-seasonal: $p = 1$, $d = 1$, $q = 1$

seasonal: $p = 0$, $d = 1$, $q = 1$

```
arima(train.ts, order=c(1,1,1), seasonal = list(order = c(0,1,1),  
  period = 12), method="ML")
```

ARIMA

- Using parameters from *auto.arima*



RMSE on validation set =78.413

ARIMA

■ Parameters

- Non-seasonal ARIMA: $\text{ARIMA}(p, d, q)$
- Seasonal ARIMA: $\text{ARIMA}(p, d, q, P, D, Q, m)$

p, P = order of the autoregressive part

d, D = degree of first differencing involved

q, Q = order of the moving average part

m = seasonal period

References

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- J. Han, M. Kamber, and J. Pei, Data Mining Concepts and Techniques, 3rd Ed., Morgan Kaufmann, 2012
- *NIST/SEMATECH e-Handbook of Statistical Methods*, <http://www.itl.nist.gov/div898/handbook/>, NIST, U.S. Department of Commerce, April 2012